

RAVEN3D: A Benchmark for 3D Structural and Relational Reasoning in Large Vision-Language Models

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Abstract

Large vision-language models (VLMs) have achieved impressive performance across a wide range of tasks, yet their ability to perform abstract visual reasoning—particularly in three-dimensional (3D) settings—remains poorly understood. In this work, we ask whether current VLMs can exhibit human-like structural and relational reasoning in zero-shot 3D analogy-making scenarios. To address this question, we introduce RAVEN3D, a large-scale benchmark comprising 108K synthesized 3D abstract visual reasoning puzzles that extends Raven’s Progressive Matrices with multiview perception, tier-based stacking layouts, and systematically designed attribute-level relations. We evaluate human participants, state-of-the-art closed-source and open-source VLMs, fine-tuned VLMs, and supervised neural networks across diverse configurations, complemented by detailed error, difficulty, and relational-effect analyses. Our results show that even the strongest large VLMs achieve on average less than half of human accuracy, with major failures stemming from limitations in attribute-level visual perception and relational induction, rather than deductive reasoning; other models perform substantially worse. These findings reveal a pronounced gap between VLMs and humans in zero-shot 3D structural and relational reasoning, highlighting critical challenges and future directions for advancing 3D abstract reasoning systems.

1. Introduction

Large vision-language models (VLMs) have recently achieved remarkable progress (OpenAI, 2023; DeepMind,

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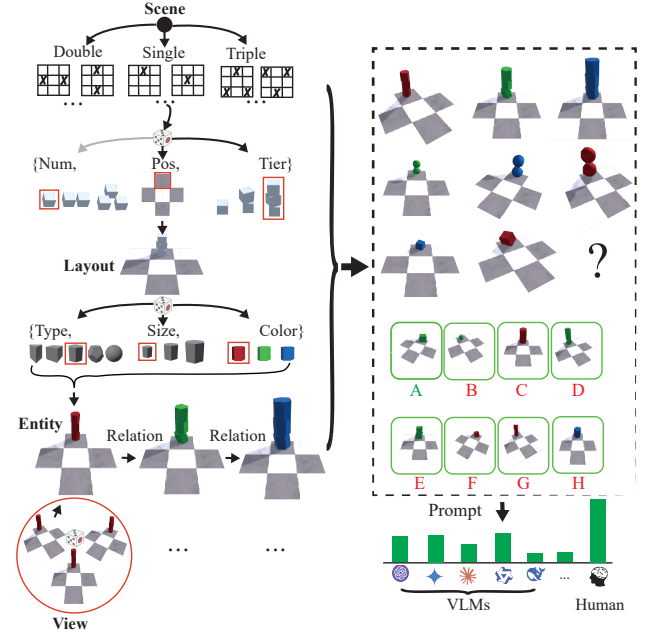


Figure 1. Overview of RAVEN3D dataset. Each puzzle contains eight context panels and eight candidate answers, generated via a four-level grammar over Scene, Component Structure (Single, Double, or Triple), Layout, and Entity, with attributes and relations applied to induce structured reasoning. Distractors are created by controlled perturbations of the correct answer, and each panel is rendered from a randomly sampled viewpoint (Front, Left, Right), enabling multi-view 3D reasoning in a zero-shot setting.

2025a; Lu et al., 2024; Zhang et al., 2024b), approaching or even surpassing human-level performance on tasks such as visual question answering, document understanding, chart reasoning, and general multimodal knowledge reasoning. Despite these advances, their reasoning behaviors still differ substantially from those of humans, revealing gaps in modeling human-like reasoning processes (Chen et al., 2025a). These limitations become particularly evident in tasks that require abstract visual reasoning (AVR), especially analogy-making and relational reasoning—core cognitive mechanisms that enable human to form abstractions and reason effectively in previously unseen scenarios (Holyoak, 2012; Hofstadter & Sander, 2013). Raven’s Progressive Matrices (RPM) (Raven, 1965) is a widely used psychometric test for relational reasoning, in which the goal is to induce an

abstract rule from a small set of examples—typically short sequences of grouped shapes—and then apply this rule to complete a new, partially observed sequence.

In recent years, the RAVEN dataset (Zhang et al., 2019a) and its variants (Hu et al., 2021; Benny et al., 2021; Małkiński & Mańdziuk, 2025) have been introduced as re-creations of Raven’s Progressive Matrices, enriching the perceptual complexity of the matrices. However, most existing work remains focused on two-dimensional (2D) abstract reasoning, with comparatively little attention devoted to 3D abstract reasoning, and only a limited number of corresponding datasets (Mondal et al., 2023). As a result, research comparing human and VLM performance on 3D visual structures generated through rule compositions remains insufficient.

Consequently, current AVR research still falls short in the following aspects: 1) Existing 3D AVR datasets are scarce and limited in scale, making them insufficient to support thorough evaluation of VLMs or subsequent optimization such as fine-tuning. 2) Prior work has predominantly focused on 3D spatial reasoning, with relatively little emphasis on structural and relational reasoning, and lacks a systematic design, evaluation, and analysis of attributes and relations. 3) Most AVR studies concentrate on supervised neural networks, reformulating AVR as a classification problem, while insufficiently exploring the zero-shot reasoning capabilities of VLMs and their comparison with supervised approaches.

Motivated by these limitations, we systematically investigate analogical reasoning in VLMs by evaluating their ability to infer and generalize structural and relational patterns. We introduce RAVEN3D (Figure 1), a synthesized 3D abstract visual reasoning benchmark, and conduct comprehensive evaluations across human participants, API-based and open-source VLMs, as well as supervised neural networks. Our evaluation results show that current VLMs struggle with analogy-making, particularly in zero-shot 3D structural and relational reasoning, revealing a substantial gap between model performance and human reasoning. Furthermore, we analyze key failure modes, examine the effects of multiview perception and fine-tuning, and provide insights into the strengths and limitations of existing approaches.

In summary, this paper proposes a novel 3D visual reasoning challenge stemming from I-RAVEN (Hu et al., 2021), which removes a bias in RAVEN’s answer sets. Our main contributions are as follows:

- **Extend Raven’s Progressive Matrices to 3D puzzles by incorporating multiview and stacking attributes.** Building on intrinsic properties of 3D polyhedral objects—such as the number of faces and tier-based stacking—we expand both entity-level attributes and layout-level relations. While preserving the canonical matrix-based configuration of puzzles, each row is rendered

from three different viewpoints, resulting in substantially increased perceptual and relational complexity compared to 2D puzzles.

- **Identify a significant gap between VLMs and human in zero-shot 3D structural and relational reasoning.** Extensive evaluations on the proposed benchmark show that VLMs struggle to generalize to unseen 3D relational configurations compared to human. We attribute this gap to distinct error patterns and difficulty sensitivities, as revealed by systematic error and difficulty analyses.
- **Evaluation the state-of-the-art VLMs and AVR models on RAVEN3D.** We curate 108,000 RAVEN3D puzzles and conduct systematic evaluations on a 3,000-instance test set, enabling comprehensive comparisons among 24 representative models, including VLMs, fine-tuned VLMs, and supervised AVR models, and revealing critical bottlenecks in current 3D structural and relational reasoning.

2. Related Work

2.1. 2D and 3D AVR Benchmarks

Abstract Visual Reasoning (AVR) tasks typically present a set of images depicting simple 2D or 3D shapes that vary systematically along specific visual attributes. These images are organized into a structured relational format—such as a sequence, grid, or matrix—that implicitly encodes a set of underlying rules. Solving an AVR task requires inferring the latent abstract rules that govern relationships among images, objects, and their attributes from observed patterns in order to generalize to the correct answer. However, existing AVR benchmarks remain limited in scope: they cover only a narrow range of reasoning patterns and rely on fixed puzzle configurations with restricted visual primitives, which may introduce evaluation biases and constrain the study of more general visual reasoning abilities (Małkiński & Mańdziuk, 2023). We review 11 representative and widely used AVR benchmark datasets, comparing them along three dimensions: shape, pattern, and type. As shown in Table 1, most existing benchmarks focus on 2D geometric shapes and reasoning over numerical, arithmetic, or temporal patterns, while only a few datasets—such as Bongard-LOGO (Nie et al., 2020) and SVRT (Fleuret et al., 2011)—are based on abstract shapes. Specifically, RAVEN (Zhang et al., 2019a), G-set (Mańdziuk & Żychowski, 2019), VAP (Hill et al., 2019), DOPT (Webb et al., 2020), and ARC (Chollet, 2019) are constructed using two-dimensional geometric shapes, whereas CLEVER-Matrices (Mondal et al., 2023), LEGO (Tang et al., 2025a), iVISPAR (Mayer et al., 2025), and the proposed RAVEN3D extend to three-dimensional settings. In addition, these benchmarks vary in their an-

swer configuration types: VAP adopts a two-row format; Bongard-LOGO and SVRT are group-based; DOPT and LEGO follow a sequential structure; and the remaining benchmarks are organized as matrices.

2.2. VLMs’ ability Evaluations

Vision-language models (VLMs), such as OpenAI’s CLIP (Radford et al., 2021), Salesforce’s BLIP (Li et al., 2022), and DeepMind’s Flamingo (Alayrac et al., 2022), have demonstrated strong reasoning and multimodal understanding capabilities, outperforming classical single-modality vision models on zero-shot classification tasks (Islam et al., 2023). With the rapid advancement of VLMs since 2022, the number of benchmarks designed to evaluate these models has grown substantially (Chia et al., 2024a; Zhang et al., 2024a; Li et al., 2025). However, most existing work evaluates models using question-answering tests or visual-spatial reasoning tasks, highlighting the necessity for more systematic evaluation protocols tailored to the distinctive capabilities of VLMs. For instance, SpatialEval (Wang et al., 2024a), Sparkle (Tang et al., 2025b) and GSR-BENCH (Rajabi & Kosecka, 2024) evaluate diverse spatial reasoning tasks using VLMs.

Unlike VQA, abstract visual reasoning (AVR) lies at the core of human reasoning (Carpenter et al., 1990; Snow et al., 1984) and reflects key aspects of high-level intelligence, particularly fluid intelligence (Jaeggi et al., 2008). Recent studies have increasingly focused on developing learning algorithms that can capture abstract visual reasoning capabilities. AVR provides both vision-level understanding and structure-level reasoning which would notably push the limit of artificial intelligence (AI) towards the center of cognitive ability test circle. While prior studies, including PuzzleVQA (Chia et al., 2024b) and PLM-RAVEN (Hu et al., 2023), show that VLMs fail to generalize effectively to simple abstract patterns, comprehensive evaluations of abstract visual reasoning—especially 3D analogical visual reasoning—remain largely unexplored for VLMs.

3. RAVEN3D

Our dataset is not a straightforward 2D-to-3D translation of existing benchmarks; rather, it fundamentally leverages the additional spatial dimension to introduce new forms of visual structure and reasoning. In particular, we model vertical stacking through an explicit Tier attribute and incorporate multi-view representations that are intrinsic to 3D perception but absent in 2D settings. As illustrated in Figure 2, RAVEN3D is organized into three primary subtasks based on object count, each further decomposed into one to three finer-grained configurations according to layout and structural variations. Compared to RAVEN (Zhang et al., 2019a), our benchmark expands the rule space from four

to five dimensions and introduces attributes unique to 3D scenes. Specifically, we extend the Layout dimension with tier-based layering and additionally require cross-view reasoning, thereby posing richer and more realistic challenges for visual analogical reasoning.

3.1. Attributes and Rules

For RAVEN3D, we adopt an attributed stochastic 3D primitive grammar, following RAVEN, with three scene configurations (Table 2). To model 3D stacking, we extend the layout with a Tier attribute, implemented via bottom-up sampling, denoted by $\mathcal{S}$, with attribute transformations governed by $\mathcal{T}$. The geometric prototypes include triangular prisms, cubes, hexagonal prisms, dodecahedra, and spheres, providing polyhedral shapes with varying face counts.

The attribute set is $A = \{\text{Number, Position, Tier, Size, Type, Color, Orientation}\}$ and the relation set is $R = \{\text{Constant, Progression, Arithmetic, Distribute Three, Shuffle}\}$. As Number and Position are strongly coupled, we introduce Shuffle to govern discrete spatial locations, while Orientation is treated as a noise attribute. Stacking and Double/Triple configurations induce multiple independently varying objects per component, resulting in an average of 5.67 attribute relations per sample. Finally, each component is rendered from a randomly selected viewpoint in $V = \{\text{Front, Left, Right}\}$, introducing structured multiview perception.

3.2. Task Definition

As illustrated in Figure 1 and formalized in Table 2, RAVEN3D extends the 2D RAVEN paradigm to 3D scenes to probe structural and relational reasoning under realistic spatial factors. Beyond replacing planar symbols with 3D geometric primitives, RAVEN3D introduces three key extensions: (1) a Tier attribute for vertical stacking and layered layouts, (2) richer scene compositions with multiple objects and layers, and (3) multi-view rendering to reflect the inherent ambiguity and occlusion in 3D perception. Based on the attributed stochastic 3D grammar in Table 2, we define three task families according to scene-level structure.

Single-Structure Reasoning contains a single 3D object with varying attributes and viewpoints, stacked into 1–3 tiers with fixed spatial positions. To explicitly control compositional depth, the 3-layer setting includes both shared and independently sampled attributes across layers, while all 1–2 layer settings adopt shared sampling.

Double-Structure Reasoning consists of two objects arranged in Left–Right or Up–Down layouts and stacked into 1–2 tiers, forming two coupled single-object structures that require disentangling parallel relational patterns.

Triple-Structure Reasoning involves scenes with up to

Table 1. Comparing the proposed RAVEN3D to related benchmarks. Our dataset employs 3D geometric shapes and emphasizes spatial structural and relational reasoning via multiview and stacking. Puzzles are arranged in matrix formats and the dataset is substantially larger than prior benchmarks. * marks aspects in which our dataset extends or differs from RAVEN and similar benchmarks.

	Dimension	2D							3D			
		RAVEN	G-set	VAP	Bongard-LOGO	SVRT	DOPT	ARC	CLEVR-Mat	LEGO	iVISPAP	RAVEN3D
Shape	Geometric	✓	✓	✓			✓	✓	✓	✓	✓	✓
	Abstract				✓	✓						
Pattern	Spatial					✓		✓		✓	✓	
	Quantities	✓	✓	✓	✓	✓	✓	✓	✓			✓
	Arithmetic	✓	✓	✓				✓	✓			✓
	Temporal	✓	✓				✓	✓	✓	✓	✓	✓
	Multiview									✓		✓*
Type	Stacking									✓		✓*
	Sequence						✓			✓		
	Two-row			✓								
	Matrix	✓	✓						✓		✓	✓
	Group				✓	✓						
	# puzzles	70,000	49,000	710,000	12,000	10,000	600,000	1,000	20,000	900	770	108,000*

Table 2. Grammar production rules used in RAVEN3D.

Level	Production Rules
Scene	Scene $\rightarrow$ Single
	Scene $\rightarrow$ Double
	Scene $\rightarrow$ Triple
Component Structure	Single $\rightarrow$ Entity $\cdot$ Layout
	Triple $\rightarrow$ Entity $\cdot$ Layout
	Double $\rightarrow$ {Left, Right}
	Double $\rightarrow$ {Up, Down}
	Left $\rightarrow$ Entity $\cdot$ Layout
	Right $\rightarrow$ Entity $\cdot$ Layout
	Up $\rightarrow$ Entity $\cdot$ Layout
	Down $\rightarrow$ Entity $\cdot$ Layout
Layout	Layer ₀ $\rightarrow$ {Number, Position}
	Layout $\rightarrow \bigcup_{i=0}^2 \mathcal{S}(\text{Layer}_0, \text{Tier}_i)$
Entity	Entity $\rightarrow$ Prototype $\cdot \mathcal{T}\{\text{Size, Type, Color}\}$
Relation	Constant
	Progression
	Arithmetic
	Distribute Three Shuffle
View	Front
	Left
	Right

three objects arranged in either linear or triangular layouts, designed to support more complex relational rules. It includes two configurations: (i) a *1-or-2-or-3* structure, where the number of objects varies under shared attribute sampling, and (ii) a *3-Shuffle* structure, which introduces an additional varying attribute (e.g., Size, Type, or Color) to enable shuffle-based relations. Together, this task family jointly

challenges reasoning over object count, spatial arrangement, and attribute interactions in 3D scenes.

Across all task types, 3-layer configurations act as a stress test for cross-layer compositional reasoning, while multiview rendering introduces realistic 3D perceptual variation. Distractors are generated following I-RAVEN (Hu et al., 2021) using attribute bisection and layout perturbation. All tasks are evaluated in a zero-shot setting for both VLMs and humans. For Double- and Triple-Structure configurations, stacking beyond two tiers would introduce severe occlusion. Therefore, except for the Single-Structure setting, the Tier attribute is limited to at most two layers.

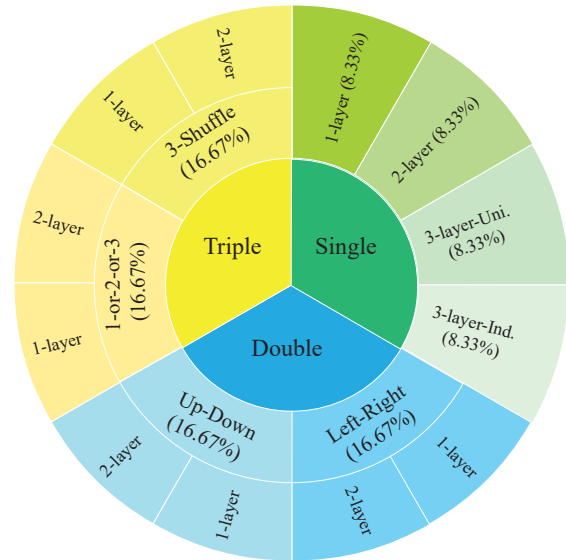


Figure 2. Problem statistics in the RAVEN3D.

4. Experimental Results

4.1. Experimental Settings

State-of-the-Art VLMs. We conduct comparisons across a broad spectrum of state-of-the-art VLMs, including both API-based and open-source models, which vary substantially in architecture, model scale, and instruction-tuning strategies. These models are selected to be representative of those most commonly evaluated in recent abstract visual reasoning (AVR) studies (Hu et al., 2023; Jiang et al., 2024). Specifically, our evaluation includes GPT-4o (OpenAI, 2024), Gemini-2.5-Pro (DeepMind, 2025b), Claude-3.5-Sonnet (Anthropic, 2024), Qwen3-VL-[4B/8B/235B] (Bai et al., 2025a), Qwen2.5-VL-[7B/32B/72B]-Instruct (Bai et al., 2025b), Qwen2-VL-2B-Instruct (Wang et al., 2024b), DeepSeek-VL2-4.5B (Wu et al., 2024), LLaVA-1.6-Vicuna-[7B/13B] (Liu et al., 2023), Emu3-8B (Wang et al., 2024c), Idefics3-8B (Laurençon et al., 2024), InternVL2.5-8B (Chen et al., 2025b), MiniCPM-V2.6-8B (Yao et al., 2024), and Pixtral-12B (Agrawal et al., 2024).

Supervised Baselines. We selected five representative networks for the RPM problem as our comparison baselines. CoPINet (Zhang et al., 2019b) models candidate answers using a contrastive reasoning module. SRAN (Hu et al., 2021) learns relational rules via hierarchical rule embeddings and gated fusion over row pairs. MRNet (Benny et al., 2021) combines multi-resolution convolutional perception with a DIST3 row/column operator for relational reasoning. PredRNet (Yang et al., 2023) employs a residual image encoder together with a predictive module to infer relations between context and candidates. HCV-ARR (He et al., 2023) uses a hierarchical ConViT with attention-based relational reasoning to jointly capture local details and global semantics.

Fine-tuned VLMs. We fine-tune Qwen3-VL-4B-Instruct on the 102K RAVEN3D training set via LoRA-based supervised fine-tuning using TRL (Werra et al., 2020) on a single NVIDIA H20 GPU for approximately 19 hours over one epoch. Compared to open-source small VLMs, fine-tuned models achieve accuracy gains of over 50 percentage points and substantially outperform larger API-based VLMs, underscoring the effectiveness of fine-tuning. Error analysis shows that these improvements primarily stem from substantial reductions in size and type errors, which dominate VLM failures prior to fine-tuning.

Dataset Difficulty. From the perspective of task difficulty, human, large and small VLMs, and supervised AVR networks exhibit markedly different behaviors. Human find the Single setting more challenging than Double and Triple, likely because the latter involve richer attribute combinations that support elimination-based reasoning, whereas Single relies on a limited set of attributes (primarily size, type, and color) requiring finer-grained comparison. In contrast,

API-based VLMs (i.e., three closed-source large VLMs and two open-source large VLMs) exhibit a different difficulty ordering, with Double being the most challenging, followed by Single, and Triple the easiest; this may reflect limited flexibility in elimination strategies, where Double increases both perceptual and rule-induction complexity, while Triple allows elimination across more attributes. Open-source smaller VLMs perform near random guessing, making difficulty analysis less informative. Supervised networks similarly find Single the most difficult, suggesting that fewer attributes provide weaker visual cues for feature-based discrimination, whereas additional attributes in Double and Triple facilitate decision-making.

Evaluation Protocol. In each experiment, we follow the standard data split protocol adopted in prior works, using 102K matrices for training, 3K for validation, and 3K for testing. To prevent models from exploiting superficial statistical correlations and to better assess their genuine reasoning ability, we intentionally introduce a distribution shift between the training and test sets while preserving the same underlying generative rules. Concretely, we modify several visual attributes during test set construction: for the type attribute, triangular prisms replace triangular pyramids and dodecahedra replace octahedra; for the color attribute, the test set uses red, green, and blue instead of the pink, blue, and yellow used in training; additionally, the size attribute is defined using different scaling ratios across the two sets.

Human Evaluation. To estimate the upper-bound performance on the Paper Folding Puzzles benchmark, we recruit 10 human subjects consisting of college students from the computer science major. In the experiments, human subjects were familiarized with solving problems using one puzzle from the Single, Double, and Triple structures respectively. After the familiarization phase, each subject was asked to complete 25 randomly selected questions from each of the 12 task configurations, ensuring comprehensive coverage of all pattern types and reasoning settings. The average accuracy across the 10 participants is reported as the human performance baseline.

4.2. Task Reasoning

Table 3 reports the performance of five API-based large VLMs, thirteen open-source smaller VLMs, five supervised neural networks, and one fine-tuned VLM, alongside human performance, across all tasks. Humans substantially outperform models: their average accuracy is more than 50 percentage points higher than API-based large VLMs and about 18 percentage points higher than fine-tuned open-source VLMs. While supervised neural networks attain performance on par with API-based large VLMs, most open-source smaller VLMs lag far behind—frequently nearing random performance—except for Qwen3-VL-[4B/8B]—

Instruct and Qwen2.5-VL-32B.

It is worth noting that the Single 3-layer setting is evaluated under two variants: shared and independent attribute sampling across layers. We find that independent sampling consistently reduces accuracy for most VLMs and supervised AVR networks, whereas shared sampling tends to facilitate reasoning. A similar pattern is observed in the Double and Triple configurations, suggesting that stacking alone does not increase task difficulty unless the number of layers reaches a certain threshold. Instead, additional layers can provide useful cues that humans and VLMs exploit to eliminate invalid candidates, while enabling supervised AVR networks to leverage richer visual differences.

4.3. Attribute Error Analysis

Figure 3 present truncated violin plots of the error analysis on different attributes. From these results, we observe that API-based large VLMs primarily make errors on size and type, while remaining relatively robust to color. Errors related to number and position are comparatively rare, and no invalid responses are observed. In contrast, open-source smaller VLMs exhibit a more uniform distribution of errors across size, type, and color, followed by a higher proportion of errors on tier, with relatively fewer errors on number and position. Notably, a substantial number of invalid errors occur, indicating weaker instruction-following capabilities. Compared to both model categories, human participants predominantly make errors on size, with comparable but lower error rates on type, color, and tier, and significantly fewer errors on number and position. From the shape of the violin plots, we observe that human errors are concentrated on a small number of attributes, resulting in a long-tailed distribution.

The five supervised neural networks achieve performance comparable to that of large VLMs. Their errors primarily occur on type and color, followed by size and tier, with relatively few errors on number and position. These results indicate that networks trained on large-scale datasets can extract effective row- and column-level features and exhibit strong visual reasoning capabilities.

We analyze the error patterns of the fine-tuned Qwen3-VL-4B-Instruct model. We find errors related to size and type are substantially reduced after fine-tuning, decreasing from approximately 20~30% to below 10%, and invalid responses are eliminated. These results offer insight into the specific factors driving the performance improvements achieved through fine-tuning.

4.4. Relational Effect Analysis

We analyze the effect of relations by comparing the performance of VLMs and supervised neural networks on a synthe-

sized dataset containing single-attribute relations with their performance on RAVEN3D, which involves multi-attribute relations. Results for API-based large VLMs show that different relations exert markedly different effects across attributes. Overall, accuracy on single-attribute relation datasets is consistently higher than on multi-attribute relation settings. Notable exceptions arise for specific attributes: for `Number` and `Size`, the `Distribute-Three` relation is substantially more challenging than other relations, while for `Type`, the `Progression` relation poses the greatest difficulty. These trends are consistent across API-based large VLMs, indicating that the observed relational effects are not model-specific. In contrast, supervised methods do not exhibit clear or consistent patterns with respect to relational effects.

The synthesized single-attribute datasets are constructed by simplifying the 2-layer configuration of the Triple 1-or-2-or-3 setting in RAVEN3D. We choose this subtask for relation comparison because its configuration is expressive enough to encompass all attribute relations, enabling a controlled and comprehensive analysis of relational effects.

4.5. Views Analysis

Unlike 2D RPM problems, our dataset incorporates multi-view perception, with each panel randomly assigned one of three viewpoints: `Front`, `Left`, or `Right`. To test whether multi-view increases visual complexity, we created a single-view variant matching the benchmark’s attributes and layouts. As shown in Figure 4, VLMs exhibit strong robustness to viewpoint variations. In contrast, Figure 5 shows that supervised AVR networks are highly sensitive to viewpoint changes, suffering an average performance drop of about 20% in the multiview setting compared to the single-view setting, with the exception of SRAN. This indicates that VLMs possess more robust visual encoders capable of handling multi-view 3D scenes, whereas supervised networks struggle to capture three-dimensional perceptual features under viewpoint changes.

4.6. Key Research Findings

1) VLMs significantly lag behind human performance in zero-shot 3D structural and relational reasoning. Even the best API-based large VLMs achieve performance that is only about half of human accuracy, while most open-source smaller VLMs operate at levels close to random guessing (1 out of 8). In contrast, human participants flexibly attend to informative attributes and systematically eliminate incorrect candidates, leading to substantially higher accuracy. Our analyses suggest that this gap arises from failures at earlier stages of the reasoning pipeline. Attribute error analysis indicates that VLMs frequently misperceive or misidentify object attributes (e.g., size, type), while relational effects analysis shows limitations in inductive reasoning, i.e.,

Table 3. Evaluating different models and human using our Raven3D dataset containing three reasoning configurations. Bold indicates the best performance. * indicates the second highest. Parenthesized numbers report the accuracy gain after fine-tuning.

	Models	Single (Acc (%))				Double (Acc (%))			Triple (Acc (%))		
		1-Layer	2-Layer	3-Layer	Avg	Left-Right	Up-Down	Avg	1-or-2-or-3	3-Shuffle	Avg
API-based	GPT-4o	37.6	39.6	38.2	38.4	32.6	28.6	30.6	39.6	44.2	41.9
	Gemini-2.5-Pro	39.6	45.2	34.6	38.5	33.6	32.6	33.1	50.2	44.4	47.3
	Claude-3.5-Sonnet	31.2	35.6	20.2	26.8	17.8	23.2	20.5	34.6	25.6	30.1
	Qwen3-VL-235B-Thinking	37.2	43.6	37.2	38.8	36.4	34.6	35.5	46.0	50.6	48.3
	Qwen2.5-VL-72B-Instruct	22.4	28.8	20.4	23.0	18.8	17.6	18.2	22.8	26.2	24.5
<13B	Qwen2-VL-2B-Instruct	9.6	10.0	11.2	10.5	12.8	10.2	11.5	9.6	10.6	10.1
	DeepSeek-VL2-4.5B	10.8	12.8	12.6	12.2	10.6	13.6	12.1	13.0	14.6	13.8
	Qwen2.5-VL-7B-Instruct	17.2	19.6	15.6	17.0	15.4	10.4	12.9	14.4	18.2	16.3
	Qwen3-VL-4B-Instruct	22.0	25.2	19.2	21.4	12.0	10.6	11.3	24.4	22.0	23.2
	LLaVA-1.6-Vicuna-7B	15.2	16.8	13.8	14.9	13.0	11.0	12.0	12.8	12.0	12.4
	Emu3-8B	9.6	12.4	11.2	11.1	10.8	10.2	10.5	10.6	7.4	9.0
	Idefics3-8B	8.4	12.0	12.6	11.4	13.6	16.2	14.9	11.4	9.8	10.6
	Qwen3-VL-8B-Instruct	27.6	32.8	26.6	28.4	17.4	18.0	19.7	30.8	32.6	31.7
	InternVL2.5-8B	15.2	11.2	15.0	14.1	14.0	15.4	14.7	13.6	11.6	12.6
	MiniCPM-V2.6-8B	13.2	12.4	10.4	11.6	13.4	13.0	13.2	15.0	12.6	13.8
>13B	LLaVA-1.6-Vicuna-13B	12.4	14.4	11.6	12.5	16.6	14.4	15.5	13.2	11.6	12.4
	Qwen2.5-VL-32B	25.6	27.6	22.0	24.3	20.2	14.8	17.5	25.2	27.0	26.1
Supervised	CoPINet	26.8	26.0	22.4	24.4	23.2	27.2	25.2	29.0	18.6	23.8
	SRAN	9.2	14.4	11.8	11.8	15.0	12.4	13.7	12.0	9.8	10.9
	MRNet	20.0	30.8	24.0	24.7	32.0	26.6	29.3	36.0	49.6	42.8
	PredRNet	23.6	42.0	30.2	31.5	40.8	42.0	41.4	34.0	54.0	44.0
	HCV-ARR	21.2	36.4	25.8	27.3	36.0	39.2	37.6	44.8	40.6	42.7
SFT	Qwen3-VL-4B-Instruct	79.2* (+57.2)	83.2* (+58.0)	70.2* (+51.0)	75.7* (+54.3)	63.4* (+51.4)	71.0* (+60.4)	67.2* (+55.9)	76.2* (+51.8)	70.2* (+48.2)	73.2* (+50.0)
	Human	82.0	90.8	84.8	85.9	92.6	94.6	93.6	92.2	92.6	92.4

inferring abstract relational rules from observed patterns. Importantly, when both attribute perception and relational induction are correct, deductive reasoning—applying the inferred rule to select the correct answer—is largely reliable for VLMs. This suggests that current VLMs are not fundamentally deficient in deduction, but are instead bottlenecked by imperfect visual perception and weak relational induction in 3D settings.

2) Improvement potential of fine-tuned VLMs and supervised AVR networks. Supervised AVR networks remain simple, efficient, and effective by casting analogical reasoning as a structured classification problem. While fine-tuned VLMs achieve substantial performance gains and improved robustness, supervised networks exhibit complementary strengths on specific visual attributes with significantly lower computational cost. The distinct error patterns and inductive biases of the two paradigms highlight their complementarity, suggesting that hybrid approaches integrating fine-tuned VLMs with structured supervised reasoning may further advance 3D analogical reasoning.

5. Conclusion and Future Work

Our dataset extends the classic two-dimensional RPM benchmarks to three-dimensional abstract visual reasoning. Evaluations on this dataset reveal that existing VLMs still exhibit substantial limitations in 3D structural and relational reasoning abilities. However, three-dimensional objects can easily occlude or interpenetrate under deep stacking, thus we avoid excessive stacking depths and complex configurations such as 2×2 or 3×3 grids in the current design. In future work, more complex configurations may be explored by incorporating interactive or dynamic human-computer interfaces, enabling richer structural compositions for more rigorous evaluations of VLM reasoning capabilities.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

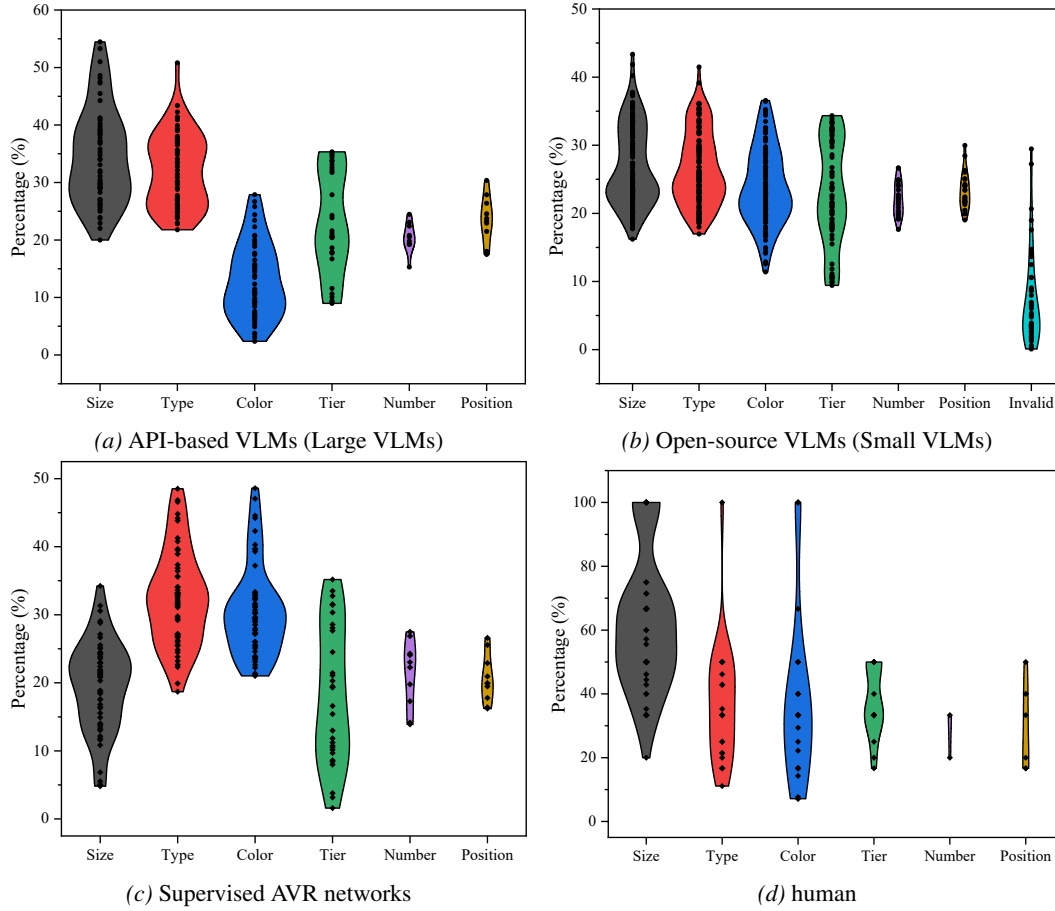


Figure 3. Error analysis of different models and human.

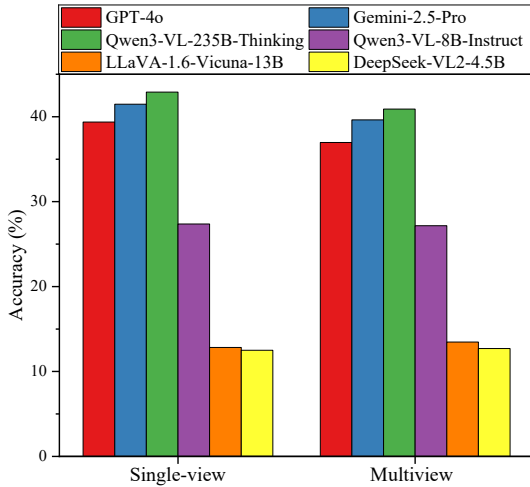


Figure 4. Single-view vs. Multiview performance of representative VLMs.

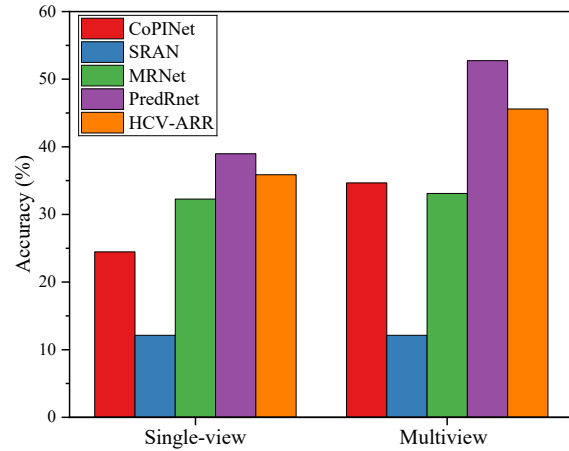


Figure 5. Single-view vs. Multiview performance of five supervised AVR networks.

References

Agrawal, P., Antoniak, S., Hanna, E. B., Bout, B., Chaplot, D., Chudnovsky, J., Costa, D., Monicault, B. D., Garg,

S., Gervet, T., Ghosh, S., Héliou, A., Jacob, P., Jiang, A. Q., Khandelwal, K., Lacroix, T., Lample, G., Casas, D. L., Lavril, T., Scao, T. L., Lo, A., Marshall, W., Martin, L., Mensch, A., Muddireddy, P., Nemychnikova, V.,

- Pellat, M., Platen, P. V., Raghuraman, N., Rozière, B., Sablayrolles, A., Saulnier, L., Sauvestre, R., Shang, W., Soletskyi, R., Stewart, L., Stock, P., Studnia, J., Subramanian, S., Vaze, S., Wang, T., and Yang, S. Pixtral 12B. *arXiv preprint arXiv:2410.07073*, 2024.
- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., Ring, R., Rutherford, E., Cabi, S., Han, T., Gong, Z., Samangooei, S., Monteiro, M., Menick, J., Borgeaud, S., Brock, A., Nematzadeh, A., Sharifzadeh, S., Binkowski, M., Barreira, R., Vinyals, O., Zisserman, A., and Simonyan, K. Flamingo: a visual language model for few-shot learning. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NeurIPS 2022)*, pp. 23716–23736, Red Hook, NY, USA, 2022. Curran Associates Inc.
- Anthropic. Claude 3.5 Sonnet, 2024.
- Bai, S., Cai, Y., Chen, R., Chen, K., Chen, X., Cheng, Z., Deng, L., Ding, W., Gao, C., Ge, C., Ge, W., Guo, Z., Huang, Q., Huang, J., Huang, F., Hui, B., Jiang, S., Li, Z., Li, M., Li, M., Li, K., Lin, Z., Lin, J., Liu, X., Liu, J., Liu, C., Liu, Y., Liu, D., Liu, S., Lu, D., Luo, R., Lv, C., Men, R., Meng, L., Ren, X., Ren, X., Song, S., Sun, Y., Tang, J., Tu, J., Wan, J., Wang, P., Wang, P., Wang, Q., Wang, Y., Xie, T., Xu, Y., Xu, H., Xu, J., Yang, Z., Yang, M., Yang, J., Yang, A., Yu, B., Zhang, F., Zhang, H., Zhang, X., Zheng, B., Zhong, H., Zhou, J., Zhou, F., Zhou, J., Zhu, Y., and Zhu, K. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025a.
- Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., and Lin, J. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025b.
- Benny, Y., Pekar, N., and Wolf, L. Scale-localized abstract reasoning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2021)*, pp. 12552–12560, Nashville, TN, 2021. IEEE.
- Carpenter, P. A., Just, M. A., and Shell, P. What one intelligence test measures: a theoretical account of the processing in the raven progressive matrices test. *Psychological review*, 97(3):404–431, 1990.
- Chen, Q., Liu, J., Qin, L., Zhang, Y., Liang, Y., Ren, S., Luan, C., Peng, D., Li, H., Guan, J., Yan, Z., Wang, J., Hu, M., Du, Y., Chen, Z., Chen, X., and Che, W. The universal landscape of human reasoning. *arXiv preprint arXiv:2510.21623*, 2025a.
- Chen, Z., Wang, W., Cao, Y., Liu, Y., Gao, Z., Cui, E., Zhu, J., Ye, S., Tian, H., Liu, Z., Gu, L., Wang, X., Li, Q., Ren, Y., Chen, Z., Luo, J., Wang, J., Jiang, T., Wang, B., He, C., Shi, B., Zhang, X., Lv, H., Wang, Y., Shao, W., Chu, P., Tu, Z., He, T., Wu, Z., Deng, H., Ge, J., Chen, K., Zhang, K., Wang, L., Dou, M., Lu, L., Zhu, X., Lu, T., Lin, D., Qiao, Y., Dai, J., and Wang, W. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271*, 2025b.
- Chia, Y. K., Hong, P., Bing, L., and Poria, S. InstructEval: Towards holistic evaluation of instruction-tuned large language models. In Miceli-Barone, A. V., Barez, F., Cohen, S., Voita, E., Hermann, U., and Lukasik, M. (eds.), *Proceedings of the First edition of the Workshop on the Scaling Behavior of Large Language Models (SCALE-LLM 2024)*, pp. 35–64, St. Julian’s, Malta, 2024a. Association for Computational Linguistics.
- Chia, Y. K., Toh, V., Ghosal, D., Bing, L., and Poria, S. PuzzleVQA: Diagnosing multimodal reasoning challenges of language models with abstract visual patterns. In Ku, L.-W., Martins, A., and Srikumar, V. (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 16259–16273, Bangkok, Thailand, 2024b. Association for Computational Linguistics.
- Chollet, F. On the measure of intelligence. *arXiv preprint arXiv:1911.01547*, 2019.
- DeepMind. Gemini: A family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2025a.
- DeepMind. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025b.
- Fleuret, F., Li, T., Dubout, C., Wampler, E. K., Yantis, S., and Geman, D. Comparing machines and humans on a visual categorization test. *Proceedings of the National Academy of Sciences*, 108(43):17621–17625, 2011.
- He, W., Zhang, J., Ren, J., Bai, R., and Jiang, X. Hierarchical convit with attention-based relational reasoner for visual analogical reasoning. In Williams, B. and Chen, Y. (eds.), *Proceedings of the 37th AAAI Conference on Artificial Intelligence (AAAI 2023)*, pp. 22–30, Washington, DC, 2023. AAAI.
- Hill, F., Santoro, A., Barrett, D. G. T., Morcos, A. S., and Lillicrap, T. Learning to make analogies by contrasting abstract relational structure. *arXiv preprint arXiv:1902.00120*, 2019.

- Hofstadter, D. R. and Sander, E. *Surfaces and essences: Analogy as the fuel and fire of thinking*. Basic Books, 2013.
- Holyoak, K. J. Analogy and relational reasoning. In Holyoak, K. J. and Morrison, R. G. (eds.), *The Oxford Handbook of Thinking and Reasoning*, chapter 13, pp. 234–259. Oxford University Press, 2012.
- Hu, S., Ma, Y., Liu, X., Wei, Y., and Bai, S. Stratified rule-aware network for abstract visual reasoning. In *Proceedings of the 35th AAAI Conference on Artificial Intelligence (AAAI 2021)*, pp. 1567–1574. AAAI, 2021.
- Hu, X., Storks, S., Lewis, R., and Chai, J. In-context analogical reasoning with pre-trained language models. In Rogers, A., Boyd-Graber, J., and Okazaki, N. (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1953–1969, Toronto, Canada, 2023. Association for Computational Linguistics.
- Islam, A., Biswas, M. R., Zaghouani, W., Belhaouari, S. B., and Shah, Z. Pushing boundaries: Exploring zero shot object classification with large multimodal models. In *2023 Tenth International Conference on Social Networks Analysis, Management and Security (SNAMS 2023)*, pp. 1–5, Abu Dhabi, United Arab Emirates, 2023. IEEE.
- Jaeggi, S. M., Buschkuhl, M., Jonides, J., and Perrig, W. J. Improving fluid intelligence with training on working memory. *Proceedings of the National Academy of Sciences*, 105(19):6829–6833, 2008.
- Jiang, Y., Zhang, J., Sun, K., Sourati, Z., Ahrabian, K., Ma, K., Ilievski, F., and Pujara, J. MARVEL: multidimensional abstraction and reasoning through visual evaluation and learning. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS 2024)*, pp. 46567–46592, Red Hook, NY, USA, 2024. Curran Associates Inc.
- Laurençon, H., Marafioti, A., Sanh, V., and Tronchon, L. Building and better understanding vision-language models: insights and future directions. *arXiv preprint arXiv:2408.12637*, 2024.
- Li, J., Li, D., Xiong, C., , and Hoi, S. BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In Chaudhuri, K. and Jegelka, S. (eds.), *International conference on machine learning (ICML 2022)*, pp. 12888–12900, Baltimore, Maryland, 2022. PMLR.
- Li, Z., Wu, X., Du, H., Liu, F., Nghiem, H., and Shi, G. A survey of state of the art large vision language models: Alignment, benchmark, evaluations and challenges. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2025)*, pp. 1578–1597, Nashville, TN, USA, 2025. IEEE.
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NeurIPS 2023)*, pp. 34892–34916, Red Hook, NY, USA, 2023. Curran Associates Inc.
- Lu, H., Liu, W., Zhang, B., Wang, B., Dong, K., Liu, B., Sun, J., Ren, T., Li, Z., Yang, H., Sun, Y., Deng, C., Xu, H., Xie, Z., and Ruan, C. Deepseek-VL: Towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525*, 2024.
- Małkiński, M. and Mańdziuk, J. A-I-RAVEN and I-RAVEN-Mesh: two new benchmarks for abstract visual reasoning. In Kwok, J. (ed.), *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2025)*, pp. 5932–5940, Montreal, Canada, 2025. AAAI.
- Mayer, J., Ballout, M., Jassim, S., Nezami, F. N., and Bruni, E. iVISPAR — an interactive visual-spatial reasoning benchmark for VLMs. In Christodoulopoulos, C., Chakraborty, T., Rose, C., and Peng, V. (eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*, pp. 26745–26769, Suzhou, China, 2025. Association for Computational Linguistics.
- Małkiński, M. and Mańdziuk, J. A review of emerging research directions in abstract visual reasoning. *Information Fusion*, 91:713–736, 2023.
- Mańdziuk, J. and Żychowski, A. DeepIQ: A human-inspired AI system for solving IQ test problems. In *2019 International Joint Conference on Neural Networks (IJCNN 2019)*, pp. 1–8, Budapest, Hungary, 2019. IEEE.
- Mondal, S. S., Webb, T., and Cohen, J. D. Learning to reason over visual objects. *arXiv preprint arXiv:2303.02260*, 2023.
- Nie, W., Yu, Z., Mao, L., Patel, A. B., Zhu, Y., and Anandkumar, A. BONGARD-LOGO: A new benchmark for human-level concept learning and reasoning. In *Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS 2020)*, pp. 16468–16480, Vancouver, Canada, 2020. Curran Associates, Inc.
- OpenAI. GPT-4V(ision) System Card, 2023.
- OpenAI. GPT-4o System Card, 2024.

- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. Learning transferable visual models from natural language supervision. In Meila, M. and Zhang, T. (eds.), *International conference on machine learning (ICML 2021)*, pp. 8748–8763, Virtual, 2021. PMLR.
- Rajabi, N. and Kosecka, J. GSR-BENCH: A benchmark for grounded spatial reasoning evaluation via multimodal LLMs. In *NeurIPS 2024 Workshop on Compositional Learning: Perspectives, Methods, and Paths Forward*, Vancouver, Canada, 2024. Curran Associates, Inc.
- Raven, J. *Advanced Progressive Matrices. Sets I and II*. London: H. K. Lewis (Distributed in the United States by The Psychological Corporation. San Antonio. TX), 1965.
- Snow, R. E., Kyllonen, P., and Marshalek, B. The topography of ability and learning correlations. *Advances in the psychology of human intelligence*, 2:47–103, 1984.
- Tang, K., Gao, J., Zeng, Y., Duan, H., Sun, Y., Xing, Z., Liu, W., Lyu, K., and Chen, K. LEGO-Puzzles: How good are MLLMs at multi-step spatial reasoning? *arXiv preprint arXiv:2503.19990*, 2025a.
- Tang, Y., Qu, A., Wang, Z., Zhuang, D., Wu, Z., Ma, W., Wang, S., Zheng, Y., Zhao, Z., and Zhao, J. Sparkle: Mastering basic spatial capabilities in vision language models elicits generalization to spatial reasoning. In Christodoulopoulos, C., Chakraborty, T., Rose, C., and Peng, V. (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 4083–4103, Suzhou, China, 2025b. Association for Computational Linguistics.
- Wang, J., Ming, Y., Shi, Z., Vineet, V., Wang, X., Li, Y., and Joshi, N. Is a picture worth a thousand words? Delving into spatial reasoning for vision language models. In *Proceedings of the 38th International Conference on Neural Information Processing Systems (NeurIPS 2024)*, pp. 75392–75421, Red Hook, NY, USA, 2024a. Curran Associates Inc.
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., Fan, Y., Dang, K., Du, M., Ren, X., Men, R., Liu, D., Zhou, C., Zhou, J., and Lin, J. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024b.
- Wang, X., Zhang, X., Luo, Z., Sun, Q., Cui, Y., Wang, J., Zhang, F., Wang, Y., Li, Z., Yu, Q., Zhao, Y., Ao, Y., Min, X., Li, T., Wu, B., Zhao, B., Zhang, B., Wang, L., Liu, G., He, Z., Yang, X., Liu, J., Lin, Y., Huang, T., and Wang, Z. Emu3: Next-token prediction is all you need. *arXiv preprint arXiv:2409.18869*, 2024c.
- Webb, T. W., Dulberg, Z., Frankland, S. M., Petrov, A. A., O’Reilly, R. C., and Cohen, J. D. Learning representations that support extrapolation. In Daumé, H. and Singh, A. P. (eds.), *Proceedings of the 37th International Conference on Machine Learning (ICML 2020)*, pp. 10136–10146. PMLR, 2020.
- Werra, L., Belkada, Y., Tunstall, L., Beeching, E., Thrush, T., Lambert, N., Huang, S., Rasul, K., and Gallouédec, Q. TRL: Transformer reinforcement learning, 2020.
- Wu, Z., Chen, X., Pan, Z., Liu, X., Liu, W., Dai, D., Gao, H., Ma, Y., Wu, C., Wang, B., Xie, Z., Wu, Y., Hu, K., Wang, J., Sun, Y., Li, Y., Piao, Y., Guan, K., Liu, A., Xie, X., You, Y., Dong, K., Yu, X., Zhang, H., Zhao, L., Wang, Y., and Ruan, C. DeepSeek-VL2: Mixture-of-experts vision-language models for advanced multimodal understanding. *arXiv preprint arXiv:2412.10302*, 2024.
- Yang, L., You, H., Zhen, Z., Wang, D., Wan, X., Xie, X., and Zhang, R. Neural prediction errors enable analogical visual reasoning in human standard intelligence tests. In Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., and Scarlett, J. (eds.), *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, pp. 39572–39583, Honolulu, Hawaii, 2023. JMLR.
- Yao, Y., Yu, T., Zhang, A., Wang, C., Cui, J., Zhu, H., Cai, T., Li, H., Zhao, W., He, Z., et al. MiniCPM-V: A GPT-4V level MLLM on your phone. *arXiv preprint arXiv:2408.01800*, 2024.
- Zhang, C., Gao, F., Jia, B., Zhu, Y., and Zhu, S. RAVEN: A dataset for relational and analogical visual reasoning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2019)*, pp. 5312–5322, Long Beach, CA, 2019a. IEEE.
- Zhang, C., Jia, B., Gao, F., Zhu, Y., Lu, H., and Zhu, S. Learning perceptual inference by contrasting. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems (NeurIPS 2019)*, pp. 1075–1087, Vancouver, Canada, 2019b. Curran Associates Inc.
- Zhang, J., Huang, J., Jin, S., and Lu, S. Vision-language models for vision tasks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(8):5625–5644, 2024a.
- Zhang, W., Cheng, Z., He, Y., Wang, M., Shen, Y., Tan, Z., Hou, G., He, M., Ma, Y., Lu, W., and Zhuang, Y. Multimodal Self-Instruct: Synthetic abstract image and visual reasoning instruction using language model. In Al-Onaizan, Y., Bansal, M., and Chen, Y.-N. (eds.), *Proceedings of the 2024 Conference on Empirical Methods in*

Natural Language Processing (EMNLP 2024), pp. 19228–19252, Miami, Florida, USA, 2024b. Association for Computational Linguistics.

A. Dataset Generation Details

The RAVEN3D dataset comprises 108K puzzles spanning three object-count configurations, three stacking depths (layers), and three viewpoints. We follow the Attributed Stochastic Image Grammar (A-SIG) framework introduced in RAVEN to represent the structured space of 3D shapes. The grammar is hierarchically decomposed into four levels: Scene, Component Structure, Layout, and Entity. Attributes are defined at both the Layout and Entity levels and are summarized in Table 4. In contrast to RAVEN, we extend the grammar with a Tier attribute that encodes the vertical stacking of 3D objects.

Table 4. Attributes in different levels of the grammar.

Level	Attributes
Layout	Number, Position, Tier
Entity	Size, Type, Color, Orientation

As shown in Table 5, we provide the relation specifications of the attributed stochastic 3D primitive grammar in Table 2. We specify relations over layout- and entity-level attributes. Size and Type support Constant, Progression, and Distribute Three (with Type progression defined over face counts). Color supports Constant and Distribute Three. Number supports Constant, Progression, Arithmetic, and Distribute Three. Position supports Constant, Distribute Three, and Shuffle. Tier supports Constant, Progression, and Arithmetic.

- **Layout attributes.**

- **Number:** the number of entities in a scene, ranging from 1 to 3.
- **Position:** each entity occupies one cell in a fixed 3×3 grid.
- **Tier:** the vertical stacking depth of entities, ranging from 1 to 3 layers.

- **Entity attributes.**

- **Type:** five primitives are used. For training, we use triangular pyramids, cubes, hexagonal prisms, octahedra, and spheres; for testing, triangular pyramids and octahedra are replaced by triangular prisms and dodecahedra, respectively.
- **Size:** three scaling factors, $\{0.9, 1.35, 1.75\}$ for training and $\{1.0, 1.45, 1.85\}$ for testing.
- **Color:** $\{\text{pink, blue, yellow}\}$ for training and $\{\text{red, green, blue}\}$ for testing.
- **Orientation:** sampled uniformly from $[-60^\circ, 60^\circ]$.

Table 5. Attribute-Level relations in RAVEN3D. C-Constant, P-Progression, A-Arithmetic, D-Distribute Three and S-Shuffle.

Configuration	Attribute	Relation
Single 1-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
Single 2-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Tier	C, P, A
Single 3-Layer-Shared	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Tier	C, P, A
Single 3-Layer-Independent	Size	C, P, D
	Type	C, P, D
	Color	C, D
Double L-R or U-D 1-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
Double L-R or U-D 2-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Tier	C, P, A
Triple 1-or-2-or-3 1-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Number	C, P, A, D
	Position	C, D
Triple 1-or-2-or-3 2-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Number	C, P, A, D
	Position	C, D
Triple 3-Shuffle 1-Layer	Tier	C, P, A
	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Position	S
Triple 3-Shuffle 2-Layer	Size	C, P, D
	Type	C, P, D
	Color	C, D
	Position	S
	Tier	C, P, A

For the Type attribute, the Progression relation is defined as a monotonic change in geometric shape complexity, ordered by the increasing number of faces.

B. Dataset Examples

We present examples with 12 different task configurations in Figure 6. By varying attributes at the Layout and Entity levels, we instantiate 12 distinct reasoning subtasks that

differ in both spatial configurations and entity attributes. These subtasks span a wide range of difficulty levels and require capabilities central to human fluid intelligence, including pattern recognition, analogy-making, and relational reasoning.

From a 3D scene perspective, our dataset introduces structured spatial information through three viewpoints—Front, Left, and Right, as multiview observation is a fundamental cue for understanding spatial relationships in 3D environments. From the perspective of geometric prototypes, we adopt common polyhedra and spheres that exhibit both symmetry and meaningful geometric variation, such as differences in the number of faces, which naturally support relations like progression. While humans can readily perceive such differences, VLMs often struggle with them, frequently confusing polyhedra with similar appearances but different face counts. To avoid confounding factors inherent to complex 3D scenes, such as occlusion and clutter, we adopt simple layouts and limited stacking depths, and refrain from using more complex configurations (e.g., 2x2 or 3x3 grids).

C. Expanded Results

C.0.1. SINGLE-VIEW VS. MULTIVIEW

Single-view renderings of 3D objects and scenes cannot capture all spatial information due to self- and inter-object occlusion, making multi-view displays more informative. To evaluate whether multi-view perception increases visual complexity for 3D structural and relational reasoning, we synthesized a dataset matching the main benchmark’s attributes and layouts but rendered in single view, and compared model accuracies in this controlled setting.

Comparing Table 3 and Table 7 with Table 6 and Table 8, we observe that except for supervised AVR networks, most models are largely insensitive to view changes. This indicates that VLMs possess robust 3D scene perception, likely due to their pretrained visual encoders effectively capturing multi-view and structural features. In contrast, supervised networks show notable performance drops under multiview conditions, reflecting limitations in their ability to extract and reason over three-dimensional visual information.

C.0.2. SINGLE-LAYER VS. MULTI-LAYER

We compare performance across different layer settings for three model categories—API-based large VLMs, open-source smaller VLMs, and supervised AVR neural networks—as shown in Figures 7 to 9. From observation, increasing the Tier attribute to two layers has little effect on reasoning difficulty and may even provide additional information that aids reasoning. However, difficulty increases when the number of layers reaches three or when attributes

are independently varied across layers. This effect is particularly pronounced in supervised AVR models, indicating that the `Tier` attribute alone has limited impact on 3D structural and relational reasoning unless the number of layers reaches a certain threshold.

C.0.3. RELATIONAL EFFECT ANALYSIS

The relation-effect analyses for other API-based large VLMs are presented in Figures 10 to 13. As shown, GPT-4o and Claude-3.5-Sonnet exhibit trends similar to Gemini-2.5-Pro on the `Number`, `Size`, and `Type` attributes, where certain relations consistently exert a stronger impact on performance than others. In contrast, the supervised baselines shown in Figures 14 to 18 do not display clear or consistent trends: each method exhibits distinct sensitivities to different attributes and relations, reflecting method-specific difficulty preferences.

The detailed experiments underlying these findings are provided here for completeness, though their main conclusions are already presented and discussed in the main text.

D. Additional Error Analysis

D.0.1. ZERO-SHOT VS. FINE-TUNE

Fine-tuned VLMs substantially outperform both API-based large VLMs and open-source small models, achieving nearly 70% accuracy. These gains are primarily driven by pronounced reductions in `Size` and `Type` errors, which dominate failures prior to fine-tuning.

To analyze which error types are mitigated by fine-tuning, we present the error analysis of Qwen3-VL-4B-Instruct before and after fine-tuning in Figures 19 and 20. Comparing the two, we observe that fine-tuning substantially reduces errors related to `Size`, `Type`, as well as invalid responses. This comparison provides clear evidence that the overall accuracy gains from fine-tuning are primarily driven by reductions in these error categories.

D.0.2. ERROR ANALYSIS ACROSS ALL TASKS

We provide error analyses under different task configurations in Tables 9 to 19, which clearly reveal which error types are more prevalent for each model across specific reasoning subtasks. These tables enable direct comparisons between models and humans, highlighting systematic differences in their respective error patterns.

Large API-based VLMs mainly fail when reasoning about object `Size` or `Type`, while handling other attributes reliably. Smaller open-source VLMs exhibit more uniformly distributed errors across `Size`, `Type`, `Color`, and `Tier`, along with a notable number of invalid responses, indicating weaker instruction-following ability. Human errors are

concentrated primarily on `Size`, yielding a long-tailed distribution over `Type` and `Color`. Supervised AVR networks achieve accuracy comparable to large VLMs, with errors dominated by `Type` and `Color`, reflecting strong visual feature extraction and reasoning capabilities.

E. Training and Fine-tuning Settings

E.0.1. AVR NETWORKS TRAINING

To ensure reproducibility and fair comparison, all supervised AVR models are trained under a unified data split and consistent evaluation protocol. We adopt a 34:1 train-validation split for all models, resulting in 102K training samples and 3K validation samples. Input images are resized to 320×320 , and all experiments are conducted with fixed random seeds.

All models are trained for 50 epochs unless early stopping is triggered, and we closely follow the original training configurations and hyperparameter settings reported in the corresponding papers or official implementations. Optimization is performed using the Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$, unless otherwise specified.

HCV-ARR is trained on the RAVEN dataset with a batch size of 8 and a learning rate of 1×10^{-3} , without weight decay or dropout. PredRNet uses a batch size of 16 and a learning rate of 1×10^{-3} , and incorporates block-level and classifier-level dropout for regularization. SRAN follows the I-RAVEN configuration, using a batch size of 4 and a learning rate of 1×10^{-4} . MRNet is trained with a batch size of 32 and a learning rate of 1×10^{-3} , employing contrastive loss with a DIST-based relational operator, multi-head reasoning, and early stopping. CoPINet is trained with a batch size of 16 and a learning rate of 1×10^{-3} , following its original implementation.

All experiments are conducted on a single NVIDIA RTX 3090 GPU with 24GB memory. By strictly following the original training settings of each method, we aim to ensure that the observed performance differences primarily reflect model capabilities rather than variations in training procedures.

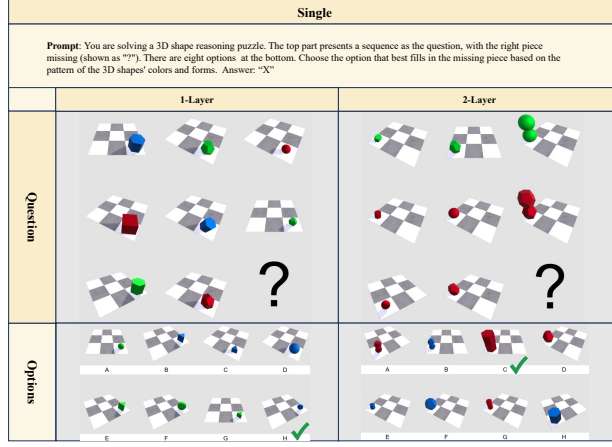
E.0.2. VLMs FINE-TUNING

We fine-tune Qwen3-VL-4B-Instruct on the RAVEN3D training set using parameter-efficient LoRA adaptation, implemented with the TRL library (Werra et al., 2020) under a supervised fine-tuning (SFT) paradigm. All experiments are conducted on a single NVIDIA H20 GPU with NVLink, and the training set contains 102K samples. To reduce memory consumption, the base model is loaded with 4-bit quantization (NF4) using bitsandbytes, while computation is performed in FP16. Fine-tuning is conducted for one full

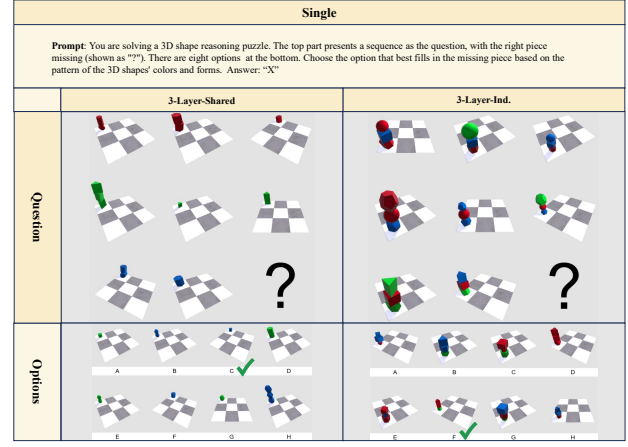
epoch, which takes approximately 18.95 hours.

LoRA is applied to the attention projection layers (q_proj , k_proj , v_proj , o_proj) with rank 32, scaling factor 32, and dropout 0.1. The model is trained for one epoch with a per-device batch size of 8 and gradient accumulation steps of 8, resulting in an effective batch size of 64. We use the AdamW 8-bit optimizer with a learning rate of 2×10^{-4} and 5 warmup steps.

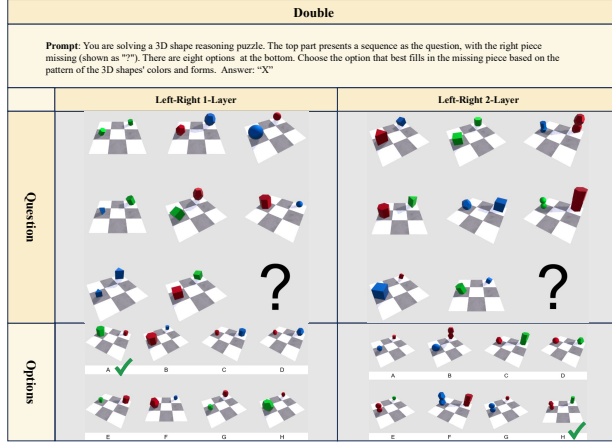
Figures 21 and 22 report the training loss curve and mean token accuracy during fine-tuning. As shown, the training loss rapidly decreases and converges to a low value within a single epoch, indicating stable optimization and effective adaptation of Qwen3-VL-4B-Instruct to the RAVEN3D reasoning task.



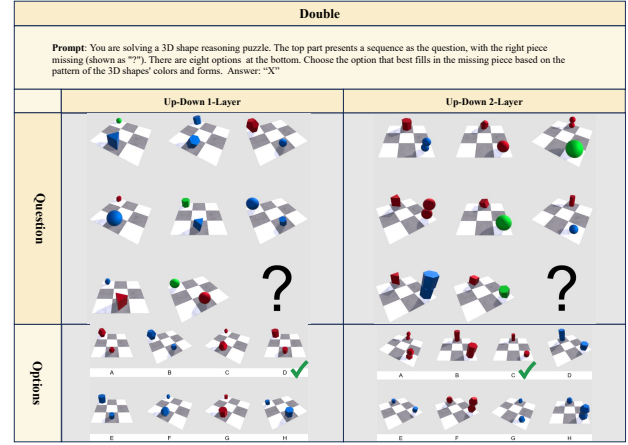
(a) Single-Structure Reasoning: 1- and 2-Layer.



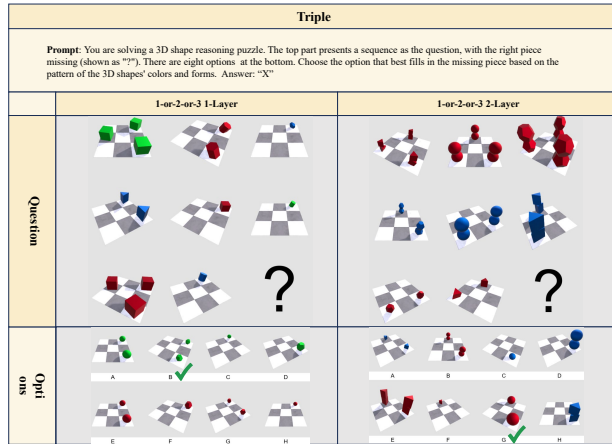
(b) Single-Structure Reasoning: 3-Layer (Shared vs. Independent Sampling).



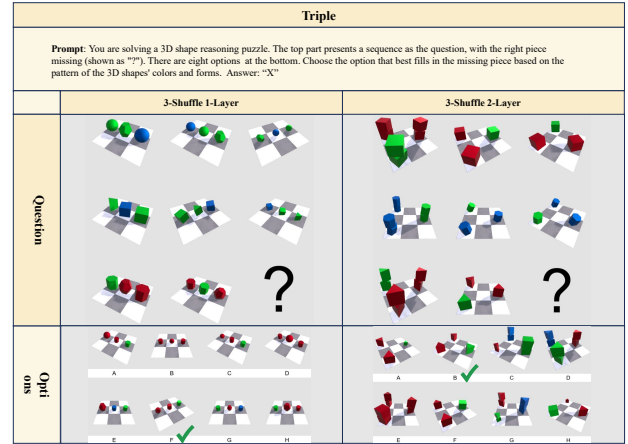
(c) Double-Structure Reasoning: 1- and 2-Layer (Left-Right).



(d) Double-Structure Reasoning: 1- and 2-Layer (Up-Down).



(e) Triple-Structure Reasoning: 1- and 2-Layer (1-or-2-or-3).



(f) Triple-Structure Reasoning: 1- and 2-Layer (3-Shuffle).

Figure 6. Examples of RAVEN3D.

Table 6. Evaluation of different layouts under specific configurations. Attribute sampling across layers is shared by default. Bold indicates the best performance. * indicates the second highest. Parenthesized numbers report the accuracy gain after fine-tuning.

	Models	Single (Acc (%))		Double (Acc (%))				Triple (Acc (%))			
		3-Layer		Left-Right		Up-Down		1-or-2-or-3		3-Shuffle	
		Shared	Ind.	1-Layer	2-Layer	1-Layer	2-Layer	1-Layer	2-Layer	1-Layer	2-Layer
API-based	GPT-4o	38.8	37.6	31.6	33.6	26.0	31.2	40.0	39.2	37.6	50.8
	Gemini-2.5-Pro	39.2	30.0	33.6	33.6	29.6	35.6	48.4	52.0	41.2	47.6
	Claude-3.5-Sonnet	23.2	17.2	18.8	16.8	22.8	23.6	32.4	36.8	27.2	24.0
	Qwen3-VL-235B-Thinking	34.0	40.4	35.2	37.6	32.4	36.8	49.6	42.4	46.4	54.8
	Qwen2.5-VL-72B-Instruct	23.6	17.2	15.2	22.4	16.8	18.4	24.4	21.2	29.2	23.2
<13B	Qwen2-VL-2B-Instruct	8.4	14.0	12.4	13.2	10.0	10.4	9.2	10.0	11.2	10.0
	DeepSeek-VL2-4.5B	13.6	11.6	10.8	10.4	13.2	14.0	17.6	8.4	13.2	16.0
	Qwen2.5-VL-7B-Instruct	20.8	10.4	16.8	14.0	10.4	10.4	15.6	13.2	18.0	18.4
	Qwen3-VL-4B-Instruct	27.2	11.2	10.8	13.2	7.2	14.0	23.2	25.6	18.8	25.2
	LLaVA-1.6-Vicuna-7B	12.8	14.8	12.8	13.2	10.8	11.2	11.6	14.0	12.8	11.2
	Emu3-8B	9.2	13.2	10.8	10.8	8.4	12.0	10.4	10.8	6.8	8.0
	Idefics3-8B	13.6	11.6	12.8	14.4	14.8	17.6	10.8	12.0	10.8	8.8
	Qwen3-VL-8B-Instruct	31.6	21.6	14.8	20.0	14.8	21.2	30.0	31.6	34.8	30.4
	InternVL2.5-8B	16.8	13.2	13.2	14.8	14.8	16.0	11.6	15.6	12.0	11.2
	MiniCPM-V2.6-8B	9.6	11.2	12.8	14.0	11.6	14.4	15.2	14.8	10.4	14.8
>13B	Pixtral-12B	14.0	15.6	17.6	15.6	16.0	13.2	16.8	10.8	15.6	12.4
	LLaVA-1.6-Vicuna-13B	11.2	12.0	15.6	17.6	13.2	15.6	10.8	15.6	10.4	12.8
Supervised	Qwen2.5-VL-32B	25.2	18.8	21.6	18.8	16.4	13.2	25.2	25.2	24.4	29.6
	CoPINet	26.8	18.0	24.4	22.0	23.6	30.8	29.6	28.4	18.8	18.4
	SRAN	12.4	11.2	17.2	12.8	12.4	12.4	12.8	11.2	10.0	9.6
	MRNet	26.0	22.0	25.6	38.4	23.6	29.6	30.8	41.2	40.0	59.2
	PredRNet	31.6	28.8	28.0	53.6	30.4	53.6	26.4	41.6	44.8	63.2
	HCV-ARR	26.4	25.2	23.6	48.4	26.4	52.0	37.2	52.4	31.6	49.6
SFT	Qwen3-VL-4B-Instruct	71.2*	69.2*	60.0*	66.8*	72.0*	70.0*	73.2*	79.2*	64.8*	75.6*
		(+44.0)	(+58.0)	(+49.2)	(+53.6)	(+64.8)	(+56.0)	(+50.0)	(+53.6)	(+46.0)	(+50.4)
	Human	82.0	87.6	90.8	94.4	93.2	96.0	90.4	94.0	92.4	92.8

Table 7. Evaluating different models and human using our synthesized single-view RAVEN3D dataset containing three reasoning configurations. Bold indicates the best performance. Parenthesized numbers report the accuracy gain after fine-tuning.

	Models	Single (Acc (%))				Double (Acc (%))			Triple (Acc (%))		
		1-Layer	2-Layer	3-Layer	Avg	Left-Right	Up-Down	Avg	1-or-2-or-3	3-Shuffle	Avg
API-based	GPT-4o	33.2	47.6	34.8	37.6	34.6	34.0	34.3	44.6	47.8	46.2
	Gemini-2.5-Pro	40.0	53.2	35.6	41.1	34.2	35.4	34.8	48.6	48.4	48.5
	Claude-3.5-Sonnet	22.8	31.2	25.2	26.1	23.2	25.8	24.5	40.2	31.4	35.8
	Qwen3-VL-235B-Thinking	37.2	43.6	37.4	38.9	38.6	40.6	39.6	45.4	55.0	50.2
	Qwen2.5-VL-72B-Instruct	20.8	25.6	23.8	23.5	20.0	15.2	17.6	25.0	25.2	25.1
<13B	Qwen2-VL-2B-Instruct	9.6	10.0	11.2	10.5	12.8	10.2	11.5	9.6	10.6	10.1
	DeepSeek-VL2-4.5B	16.4	11.6	11.6	12.8	15.0	9.8	12.4	11.0	13.6	12.3
	Qwen2.5-VL-7B-Instruct	14.0	13.2	16.2	14.9	13.8	14.4	14.1	14.6	19.0	16.8
	Qwen3-VL-4B-Instruct	25.2	26.0	15.2	20.4	12.6	13.8	13.2	20.0	25.2	22.6
	LLaVA-1.6-Vicuna-7B	10.8	12.8	14.8	13.3	12.4	11.2	11.8	10.6	12.8	11.7
	Emu3-8B	9.6	14.4	11.4	11.7	12.8	11.0	11.9	10.0	7.6	8.8
	Idefics3-8B	9.6	11.2	12.6	11.5	14.0	12.2	13.1	12.8	8.8	10.8
	Qwen3-VL-8B-Instruct	32.8	31.6	25.8	29.0	20.6	18.8	19.7	30.4	36.4	33.4
	InternVL2.5-8B	15.2	12.4	13.6	13.7	12.8	10.8	11.8	17.0	12.2	14.6
	MiniCPM-V2.6-8B	11.2	10.8	13.0	12.0	12.6	15.0	13.8	9.6	12.0	10.8
	Pixtral-12B	17.6	18.4	11.8	14.9	13.8	16.2	15.0	14.6	14.0	14.3
>13B	LLaVA-1.6-Vicuna-13B	14.4	10.8	11.8	12.2	12.8	14.4	13.6	13.4	12.0	12.7
	Qwen2.5-VL-32B	25.2	23.6	22.4	23.4	16.4	18.8	17.6	30.8	32.2	31.5
Supervised	CoPINet	22.8	28.8	29.0	27.4	32.2	36.0	34.1	42.8	42.2	42.5
	SRAN	9.2	14.4	11.8	11.8	15.0	12.4	13.7	12.0	9.8	10.9
	MRNet	24.8	32.8	23.0	25.9	26.4	24.0	25.2	35.0	61.4	48.2
	PredRNet	33.2	57.2	40.8	43.0	59.6	55.2	57.4	46.0	69.6	57.8
	HCV-ARR	35.6	51.2	36.0	39.7	45.0	46.6	45.8	49.2	53.4	51.3
SFT	Qwen3-VL-4B-Instruct	77.2 (+52.0)	84.0 (+58.0)	69.4 (+54.2)	75.0 (+54.6)	65.0 (+52.4)	73.6 (+59.8)	69.3 (+56.1)	77.0 (+57.0)	69.6 (+44.4)	73.3 (+50.7)

Table 8. Evaluation of different layouts under specific configurations on our synthesized single-view RAVEN3D dataset. Bold indicates the best performance. Parenthesized numbers report the accuracy gain after fine-tuning.

	Models	Single (Acc (%))		Double (Acc (%))				Triple (Acc (%))			
		3-Layer		Left-Right		Up-Down		1-or-2-or-3		3-Shuffle	
		Uni.	Ind.	1-Layer	2-Layer	1-Layer	2-Layer	1-Layer	2-Layer	1-Layer	2-Layer
API-based	GPT-4o	37.2	32.4	32.8	36.4	29.6	38.4	45.2	44.0	48.0	47.6
	Gemini-2.5-Pro	40.8	30.4	34.8	33.6	32.4	38.4	48.8	48.4	45.6	51.2
	Claude-3.5-Sonnet	31.2	19.2	24.0	22.4	23.6	28.0	36.4	44.0	30.8	32.0
	Qwen3-VL-235B-Thinking	32.0	42.4	37.6	39.6	38.4	42.8	44.8	46.0	54.8	55.2
	Qwen2.5-VL-72B-Instruct	28.8	18.8	18.4	21.6	11.6	18.8	25.2	24.8	24.4	26.0
<13B	Qwen2-VL-2B-Instruct	8.4	14.0	12.4	13.2	10.0	10.4	9.2	10.0	11.2	10.0
	DeepSeek-VL2-4.5B	9.6	13.6	14.4	15.6	6.4	13.2	14.4	7.6	12.0	15.2
	Qwen2.5-VL-7B-Instruct	16.4	16	12.8	14.8	15.2	13.6	16.4	12.8	18.4	19.6
	Qwen3-VL-4B-Instruct	21.2	9.2	10.4	14.8	9.2	18.4	18.8	21.2	22.4	28.0
	LLaVA-1.6-Vicuna-7B	12.8	16.8	12.0	12.8	11.6	10.8	9.2	12.0	12.0	13.6
	Emu3-8B	9.2	13.6	14.4	11.2	9.6	12.4	7.6	12.4	8.0	7.2
	Idefics3-8B	14.0	11.2	13.2	14.8	12.4	12.0	12.0	13.6	8.8	8.8
	Qwen3-VL-8B-Instruct	32.8	31.6	20.4	20.8	14.4	23.2	34.0	26.8	39.6	33.2
	InternVL2.5-8B	16.0	11.2	12.0	13.6	12.0	9.6	13.2	20.8	11.6	12.8
	MiniCPM-V2.6-8B	12.8	13.2	12.4	12.8	16.8	13.2	8.8	10.4	12.4	11.6
>13B	Pixtral-12B	10.8	12.8	12.8	14.8	16.4	16.0	15.2	14.0	13.6	14.4
	LLaVA-1.6-Vicuna-13B	13.6	10.0	10.8	14.8	15.6	13.2	16.0	10.8	10.4	13.6
	Qwen2.5-VL-32B	26.4	18.4	19.6	13.2	18.4	19.2	28.4	33.2	31.2	33.2
Supervised	CoPINet	29.6	28.4	28.8	35.6	31.6	40.4	41.2	44.4	36.0	48.4
	SRAN	12.4	11.2	17.2	12.8	12.4	12.4	12.8	11.2	10.0	9.6
	MRNet	22.4	23.6	23.2	29.6	22.0	26.0	32.8	37.2	58.0	64.8
	PredRNet	35.6	46.0	46.8	72.4	43.2	67.2	39.6	52.4	63.2	76.0
	HCV-ARR	34.8	37.2	32.4	57.6	33.6	59.6	40.0	58.4	44.8	62.0
SFT	Qwen3-VL-4B-Instruct	71.6 (+50.4)	67.2 (+58.0)	61.6 (+51.2)	68.4 (+53.6)	76.0 (+66.8)	71.2 (+52.8)	75.2 (+56.4)	78.8 (+57.6)	65.2 (+42.8)	74.0 (+46.0)

Table 9. Error analysis for Single 1-Layer configuration on the RAVEN3D Dataset.

	Models	Single (1-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	51.01%	41.41%	7.58%	-/-	-/-	-/-	-/-
	Gemini-2.5-Pro	44.26%	50.82%	4.92%	-/-	-/-	-/-	-/-
	Claude-3.5-Sonnet	48.62%	39.91%	11.47%	-/-	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	54.46%	37.13%	8.42%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	40.21%	40.91%	18.88%	-/-	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	22.39%	24.85%	23.31%	-/-	-/-	-/-	29.45%
	DeepSeek-VL2-4.5B	35.48%	32.65%	31.88%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	37.25%	34.45%	28.29%	-/-	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	40.21%	39.15%	16.73%	-/-	-/-	-/-	3.91%
	LLaVA-1.6-Vicuna-7B	33.06%	32.53%	34.41%	-/-	-/-	-/-	-/-
	Emu3-8B	33.69%	29.18%	29.18%	-/-	-/-	-/-	7.96%
	Idefics3-8B	31.28%	36.15%	32.56%	-/-	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	43.35%	35.36%	19.77%	-/-	-/-	-/-	1.52%
	InternVL2.5-8B	35.16%	36.02%	28.82%	-/-	-/-	-/-	-/-
	MiniCPM-V2.6-8B	31.93%	34.56%	33.51%	-/-	-/-	-/-	-/-
≥13B	Pixtral-12B	36.23%	35.33%	28.44%	-/-	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	30.32%	35.11%	34.57%	-/-	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	41.85%	41.48%	16.67%	-/-	-/-	-/-	-/-
	CoPINet	4.82%	46.59%	48.59%	-/-	-/-	-/-	-/-
	SRAN	31.33%	37.34%	31.33%	-/-	-/-	-/-	-/-
	MRNet	27.13%	35.65%	37.22%	-/-	-/-	-/-	-/-
	PredRNet	26.67%	44.81%	28.52%	-/-	-/-	-/-	-/-
SFT	HCV-ARR	15.55%	44.17%	40.28%	-/-	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	3.70%	31.48%	64.81%	-/-	-/-	-/-	-/-
	Human	64.81%	24.07%	11.11%	-/-	-/-	-/-	-/-

Table 10. Error analysis for Single 2-Layer configuration on the RAVEN3D Dataset.

	Models	Single (2-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	36.79%	33.57%	5.36%	24.29%	-/-	-/-	-/-
	Gemini-2.5-Pro	38.29%	35.14%	5.86%	20.72%	-/-	-/-	-/-
	Claude-3.5-Sonnet	29.74%	35.62%	10.78%	23.86%	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	38.82%	36.71%	6.75%	17.72%	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	31.09%	30.21%	10.85%	27.86%	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	21.14%	24.09%	18.64%	22.50%	-/-	-/-	13.64%
	DeepSeek-VL2-4.5B	24.52%	25.10%	24.32%	26.06%	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	28.31%	26.29%	20.00%	25.39%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	24.93%	29.54%	20.60%	21.41%	-/-	-/-	3.52%
	LLaVA-1.6-Vicuna-7B	23.29%	26.07%	26.71%	23.93%	-/-	-/-	-/-
	Emu3-8B	23.74%	26.81%	23.08%	22.64%	-/-	-/-	3.74%
	Idefics3-8B	23.15%	25.10%	26.07%	25.68%	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	30.56%	29.32%	16.05%	23.46%	-/-	-/-	0.62%
	InternVL2.5-8B	24.59%	27.89%	21.49%	26.03%	-/-	-/-	-/-
	MiniCPM-V2.6-8B	26.76%	23.83%	25.00%	24.41%	-/-	-/-	-/-
≥13B	Pixtral-12B	27.25%	27.69%	21.32%	23.74%	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	23.80%	25.80%	25.00%	25.40%	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	34.22%	24.06%	14.97%	26.74%	-/-	-/-	-/-
	CoPINet	6.86%	31.71%	32.86%	28.57%	-/-	-/-	-/-
	SRAN	24.52%	27.10%	23.87%	24.52%	-/-	-/-	-/-
	MRNet	23.58%	31.45%	29.56%	15.41%	-/-	-/-	-/-
	PredRNet	13.36%	39.68%	30.36%	16.60%	-/-	-/-	-/-
SFT	HCV-ARR	17.60%	34.08%	39.70%	8.61%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	9.52%	20.63%	46.03%	23.81%	-/-	-/-	-/-
	Human	51.72%	24.14%	6.90%	17.24%	-/-	-/-	-/-

Table 11. Error analysis for Single 3-Layer configuration on the RAVEN3D Dataset.

	Models	Single (3-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	35.83%	34.10%	20.00%	10.07%	-/-	-/-	-/-
	Gemini-2.5-Pro	34.76%	35.59%	20.69%	8.97%	-/-	-/-	-/-
	Claude-3.5-Sonnet	34.01%	33.37%	23.42%	9.20%	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	37.22%	35.6%	16.54%	10.64%	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	34.04%	32.15%	24.39%	9.42%	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	28.65%	27.60%	27.50%	9.90%	-/-	-/-	6.35%
	DeepSeek-VL2-4.5B	30.49%	29.59%	29.32%	10.61%	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	29.89%	30.17%	29.7%	10.25%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	29.01%	28.65%	22.92%	10.53%	-/-	-/-	8.89%
	LLaVA-1.6-Vicuna-7B	28.78%	29.8%	28.87%	12.55%	-/-	-/-	-/-
	Emu3-8B	28.25%	28.25%	29.00%	10.87%	-/-	-/-	3.62%
	Idefics3-8B	29.21%	29.03%	30.63%	11.13%	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	32.15%	31.69%	23.65%	10.68%	-/-	-/-	1.84%
	InternVL2.5-8B	31.13%	29.69%	29.31%	9.87%	-/-	-/-	-/-
	MiniCPM-V2.6-8B	29.51%	29.68%	28.89%	11.84%	-/-	-/-	0.09%
>=13B	Pixtral-12B	31.14%	30.66%	27.60%	10.60%	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	28.38%	28.29%	30.62%	12.71%	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	33.48%	31.71%	25.39%	9.42%	-/-	-/-	-/-
	CoPINet	22.89%	33.08%	33.30%	10.74%	-/-	-/-	-/-
	SRAN	28.80%	29.77%	30.21%	11.22%	-/-	-/-	-/-
	MRNet	29.07%	31.1%	31.42%	8.41%	-/-	-/-	-/-
	PredRNet	25.55%	33.62%	30.68%	10.15%	-/-	-/-	-/-
SFT	HCV-ARR	24.34%	32.64%	33.33%	9.69%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	26.14%	29.79%	34.35%	9.73%	-/-	-/-	-/-
	Human	60.00%	18.33%	3.33%	18.33%	-/-	-/-	-/-

Table 12. Error analysis for Left-Right 1-Layer configuration on the RAVEN3D Dataset.

	Models	Double (Left-Right 1-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	53.28%	37.55%	9.17%	-/-	-/-	-/-	-/-
	Gemini-2.5-Pro	40.85%	43.40%	15.74%	-/-	-/-	-/-	-/-
	Claude-3.5-Sonnet	37.10%	37.10%	25.81%	-/-	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	48.18%	42.27%	9.55%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	41.21%	30.91%	27.88%	-/-	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	31.25%	25.30%	25.89%	-/-	-/-	-/-	17.56%
	DeepSeek-VL2-4.5B	34.36%	33.33%	32.31%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	37.57%	32.66%	29.77%	-/-	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	26.75%	25.84%	26.75%	-/-	-/-	-/-	20.67%
	LLaVA-1.6-Vicuna-7B	33.52%	34.64%	31.84%	-/-	-/-	-/-	-/-
	Emu3-8B	28.99%	28.70%	28.12%	-/-	-/-	-/-	14.20%
	Idefics3-8B	32.80%	34.67%	32.53%	-/-	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	36.29%	32.86%	27.43%	-/-	-/-	-/-	3.43%
	InternVL2.5-8B	30.19%	33.24%	36.57%	-/-	-/-	-/-	-/-
	MiniCPM-V2.6-8B	31.65%	33.51%	34.84%	-/-	-/-	-/-	-/-
≥13B	Pixtral-12B	33.72%	33.72%	32.56%	-/-	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	30.98%	35.33%	33.70%	-/-	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	35.83%	34.85%	29.32%	-/-	-/-	-/-	-/-
	CoPINet	16.43%	41.26%	42.31%	-/-	-/-	-/-	-/-
	SRAN	30.57%	36.86%	32.57%	-/-	-/-	-/-	-/-
	MRNet	21.79%	38.93%	39.29%	-/-	-/-	-/-	-/-
	PredRNet	23.81%	46.83%	29.37%	-/-	-/-	-/-	-/-
	HCV-ARR	16.43%	41.26%	42.31%	-/-	-/-	-/-	-/-
SFT	Qwen3-VL-4B-Instruct	9.17%	24.77%	66.06%	-/-	-/-	-/-	-/-
	Human	79.17%	12.50%	8.33%	-/-	-/-	-/-	-/-

Table 13. Error analysis for Left-Right 2-Layer configuration on the RAVEN3D Dataset.

	Models	Double (Left-Right 2-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	26.21%	29.06%	9.40%	35.33%	-/-	-/-	-/-
	Gemini-2.5-Pro	30.16%	26.90%	9.24%	33.70%	-/-	-/-	-/-
	Claude-3.5-Sonnet	26.64%	25.79%	15.43%	32.14%	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	31.75%	29.67%	6.82%	31.75%	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	23.77%	23.77%	17.49%	34.98%	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	19.50%	22.37%	24.28%	30.59%	-/-	-/-	3.25%
	DeepSeek-VL2-4.5B	22.55%	22.03%	22.9%	32.52%	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	23.36%	24.13%	20.66%	31.85%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	22.41%	19.18%	19.61%	28.23%	-/-	-/-	10.56%
	LLaVA-1.6-Vicuna-7B	23.50%	20.49%	22.74%	33.27%	-/-	-/-	-/-
	Emu3-8B	21.40%	23.06%	21.59%	30.81%	-/-	-/-	3.14%
	Idefics3-8B	21.66%	22.92%	22.74%	32.67%	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	22.55%	24.84%	17.75%	32.15%	-/-	-/-	2.71%
	InternVL2.5-8B	20.52%	25.19%	21.83%	32.46%	-/-	-/-	-/-
	MiniCPM-V2.6-8B	22.62%	23.18%	22.62%	31.59%	-/-	-/-	-/-
≥13B	Pixtral-12B	23.02%	24.15%	20.57%	32.26%	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	23.06%	21.90%	24.03%	31.01%	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	24.60%	23.99%	19.15%	32.26%	-/-	-/-	-/-
	CoPINet	10.84%	26.77%	27.21%	35.18%	-/-	-/-	-/-
	SRAN	22.40%	23.84%	22.22%	31.54%	-/-	-/-	-/-
	MRNet	21.39%	26.67%	23.89%	28.06%	-/-	-/-	-/-
	PredRNet	13.68%	32.91%	23.08%	30.34%	-/-	-/-	-/-
SFT	HCV-ARR	16.80%	33.20%	28.91%	21.09%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	15.38%	17.16%	33.14%	34.32%	-/-	-/-	-/-
	Human	47.83%	13.04%	8.70%	30.43%	-/-	-/-	-/-

Table 14. Error analysis for Up-Down 1-Layer configuration on the RAVEN3D Dataset.

	Models	Double (Up-Down 1-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	47.49%	39.00%	13.51%	-/-	-/-	-/-	-/-
	Gemini-2.5-Pro	47.30%	36.93%	15.77%	-/-	-/-	-/-	-/-
	Claude-3.5-Sonnet	41.22%	36.49%	22.30%	-/-	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	45.49%	39.34%	15.16%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	35.28%	38.04%	26.69%	-/-	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	23.15%	29.08%	28.78%	-/-	-/-	-/-	18.99%
	DeepSeek-VL2-4.5B	35.31%	29.92%	34.77%	-/-	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	33.25%	36.08%	30.67%	-/-	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	25.70%	25.08%	21.98%	-/-	-/-	-/-	27.24%
	LLaVA-1.6-Vicuna-7B	31.65%	31.91%	36.44%	-/-	-/-	-/-	-/-
	Emu3-8B	32.68%	29.05%	27.65%	-/-	-/-	-/-	10.61%
	Idefics3-8B	35.73%	32.00%	32.27%	-/-	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	37.76%	29.31%	26.89%	-/-	-/-	-/-	6.04%
	InternVL2.5-8B	34.59%	36.05%	29.36%	-/-	-/-	-/-	-/-
	MiniCPM-V2.6-8B	34.15%	30.62%	35.23%	-/-	-/-	-/-	-/-
≥13B	Pixtral-12B	33.52%	35.44%	31.04%	-/-	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	34.07%	30.75%	35.18%	-/-	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	34.57%	33.64%	31.79%	-/-	-/-	-/-	-/-
	CoPINet	11.61%	43.82%	44.57%	-/-	-/-	-/-	-/-
	SRAN	34.23%	32.88%	32.88%	-/-	-/-	-/-	-/-
	MRNet	20.88%	39.56%	39.56%	-/-	-/-	-/-	-/-
	PredRNet	21.94%	48.52%	29.54%	-/-	-/-	-/-	-/-
	HCV-ARR	16.30%	39.49%	44.20%	-/-	-/-	-/-	-/-
SFT	Qwen3-VL-4B-Instruct	14.29%	20.78%	64.94%	-/-	-/-	-/-	-/-
	Human	66.67%	11.11%	22.22%	-/-	-/-	-/-	-/-

Table 15. Error analysis for Up-Down 2-Layer configuration on the RAVEN3D Dataset.

	Models	Double (Up-Down 2-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	26.13%	28.80%	10.67%	34.40%	-/-	-/-	-/-
	Gemini-2.5-Pro	29.06%	25.07%	11.11%	34.76%	-/-	-/-	-/-
	Claude-3.5-Sonnet	25.76%	24.89%	15.50%	33.84%	-/-	-/-	-/-
	Qwen3-VL-235B-Thinking	29.81%	26.18%	11.42%	32.59%	-/-	-/-	-/-
	Qwen2.5-VL-72B-Instruct	22.02%	24.04%	20.81%	33.13%	-/-	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	22.32%	22.14%	23.05%	29.76%	-/-	-/-	2.72%
	DeepSeek-VL2-4.5B	22.78%	22.42%	22.6%	32.21%	-/-	-/-	-/-
	Qwen2.5-VL-7B-Instruct	25.28%	22.14%	21.22%	31.37%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	19.78%	23.01%	20.00%	28.60%	-/-	-/-	8.60%
	LLaVA-1.6-Vicuna-7B	24.64%	23.56%	21.22%	30.58%	-/-	-/-	-/-
	Emu3-8B	20.49%	23.34%	22.01%	30.55%	-/-	-/-	3.61%
	Idefics3-8B	22.14%	22.33%	23.11%	32.43%	-/-	-/-	-/-
	Qwen3-VL-8B-Instruct	23.17%	22.34%	19.42%	33.19%	-/-	-/-	1.88%
	InternVL2.5-8B	23.21%	22.22%	20.63%	33.93%	-/-	-/-	-/-
	MiniCPM-V2.6-8B	23.06%	26.20%	20.48%	30.26%	-/-	-/-	-/-
>=13B	Pixtral-12B	23.29%	20.89%	22.37%	33.46%	-/-	-/-	-/-
	LLaVA-1.6-Vicuna-13B	22.08%	21.70%	23.40%	32.83%	-/-	-/-	-/-
Supervised	Qwen2.5-VL-32B	24.43%	20.99%	20.23%	34.35%	-/-	-/-	-/-
	CoPINet	13.10%	25.44%	27.96%	33.50%	-/-	-/-	-/-
	SRAN	21.16%	22.66%	23.41%	32.77%	-/-	-/-	-/-
	MRNet	18.81%	25.99%	23.76%	31.44%	-/-	-/-	-/-
	PredRNet	14.04%	33.19%	25.11%	27.66%	-/-	-/-	-/-
SFT	HCV-ARR	13.84%	31.70%	33.04%	21.43%	-/-	-/-	-/-
	Qwen3-VL-4B-Instruct	10.67%	14.67%	36.00%	38.67%	-/-	-/-	-/-
	Human	62.5%	6.25%	-/-	31.25%	-/-	-/-	-/-

Table 16. Error analysis for Triple 1-or-2-or-3 1-Layer configuration on the RAVEN3D Dataset.

	Models	Triple (1-or-2-or-3 1-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	39.02%	37.50%	3.79%	-/-	19.32%	-/-	-/-
	Gemini-2.5-Pro	38.46%	34.84%	3.62%	-/-	23.08%	-/-	-/-
	Claude-3.5-Sonnet	30.46%	31.46%	17.55%	-/-	20.53%	-/-	-/-
	Qwen3-VL-235B-Thinking	32.06%	37.80%	14.83%	-/-	15.31%	-/-	-/-
	Qwen2.5-VL-72B-Instruct	25.00%	32.80%	17.74%	-/-	24.46%	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	23.23%	21.29%	20.65%	-/-	22.37%	-/-	12.47%
	DeepSeek-VL2-4.5B	22.80%	25.59%	26.67%	-/-	24.95%	-/-	-/-
	Qwen2.5-VL-7B-Instruct	25.54%	27.47%	22.96%	-/-	24.03%	-/-	-/-
	Qwen3-VL-4B-Instruct	26.20%	29.14%	18.72%	-/-	20.86%	-/-	5.08%
	LLaVA-1.6-Vicuna-7B	23.09%	25.24%	26.81%	-/-	24.85%	-/-	-/-
	Emu3-8B	23.37%	23.37%	24.21%	-/-	22.11%	-/-	6.95%
	Idefics3-8B	23.98%	24.76%	26.50%	-/-	24.76%	-/-	-/-
	Qwen3-VL-8B-Instruct	30.03%	32.07%	14.87%	-/-	22.74%	-/-	0.29%
	InternVL2.5-8B	27.39%	24.84%	23.57%	-/-	24.20%	-/-	-/-
	MiniCPM-V2.6-8B	23.33%	26.77%	25.56%	-/-	24.14%	-/-	0.20%
≥13B	Pixtral-12B	27.27%	23.14%	22.93%	-/-	26.65%	-/-	-/-
	LLaVA-1.6-Vicuna-13B	23.79%	25.81%	23.79%	-/-	26.61%	-/-	-/-
Supervised	Qwen2.5-VL-32B	32.35%	27.49%	16.71%	-/-	23.45%	-/-	-/-
	CoPINet	5.07%	36.42%	31.64%	-/-	26.87%	-/-	-/-
	SRAN	25.05%	24.85%	26.04%	-/-	24.06%	-/-	-/-
	MRNet	21.75%	32.02%	32.33%	-/-	13.9%	-/-	-/-
	PredRNet	17.52%	32.35%	22.64%	-/-	27.49%	-/-	-/-
	HCV-ARR	14.92%	35.59%	32.2%	-/-	17.29%	-/-	-/-
SFT	Qwen3-VL-4B-Instruct	0.94%	16.98%	48.11%	-/-	33.96%	-/-	-/-
	Human	64.52%	9.68%	9.68%	-/-	16.13%	-/-	-/-

Table 17. Error analysis for Triple 1-or-2-or-3 2-Layer configuration on the RAVEN3D Dataset.

	Models	Triple (1-or-2-or-3 2-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	29.29%	26.92%	2.37%	21.60%	19.82%	-/-	
	Gemini-2.5-Pro	28.90%	27.76%	3.04%	17.87%	22.43%	-/-	
	Claude-3.5-Sonnet	22.92%	27.34%	9.9%	20.57%	19.27%	-/-	-/-
	Qwen3-VL-235B-Thinking	27.02%	27.02%	6.52%	18.63%	20.81%	-/-	-/-
	Qwen2.5-VL-72B-Instruct	20.00%	25.85%	13.77%	21.13%	19.25%	-/-	-/-
<13B	Qwen2-VL-2B-Instruct	19.86%	18.65%	16.75%	17.96%	21.42%	-/-	5.35%
	DeepSeek-VL2-4.5B	21.67%	19.04%	19.66%	20.43%	19.20%	-/-	-/-
	Qwen2.5-VL-7B-Instruct	18.84%	21.65%	18.68%	21.65%	19.17%	-/-	-/-
	Qwen3-VL-4B-Instruct	21.11%	21.93%	18.03%	16.80%	19.88%	-/-	2.25%
	LLaVA-1.6-Vicuna-7B	18.26%	19.08%	19.74%	21.05%	21.88%	-/-	-/-
	Emu3-8B	20.19%	18.78%	21.60%	17.53%	20.50%	-/-	1.41%
	Idefics3-8B	20.26%	17.99%	22.04%	18.31%	21.39%	-/-	-/-
	Qwen3-VL-8B-Instruct	25.00%	22.50%	14.25%	17.75%	19.25%	-/-	1.25%
	InternVL2.5-8B	18.66%	21.85%	19.66%	22.18%	17.65%	-/-	-/-
	MiniCPM-V2.6-8B	19.29%	22.51%	19.61%	18.49%	20.10%	-/-	-/-
>=13B	Pixtral-12B	20.99%	19.97%	17.24%	21.50%	20.31%	-/-	-/-
	LLaVA-1.6-Vicuna-13B	19.61%	19.77%	19.94%	20.74%	19.94%	-/-	-/-
Supervised	Qwen2.5-VL-32B	24.34%	22.29%	12.88%	20.86%	19.63%	-/-	-/-
	CoPINet	5.53%	25.58%	25.58%	20.28%	23.04%	-/-	-/-
	SRAN	21.17%	18.7%	21.02%	19.32%	19.78%	-/-	-/-
	MRNet	23.01%	24.48%	25.37%	12.98%	14.16%	-/-	-/-
	PredRNet	11.82%	31.31%	24.6%	7.99%	24.28%	-/-	-/-
SFT	HCV-ARR	13.03%	33.19%	27.73%	3.78%	22.27%	-/-	-/-
	Qwen3-VL-4B-Instruct	1.94%	11.65%	38.83%	19.42%	28.16%	-/-	-/-
	Human	45.83%	20.83%	16.67%	4.17%	12.5%	-/-	-/-

Table 18. Error analysis for Triple 3-Shuffle 1-Layer configuration on the RAVEN3D Dataset.

	Models	Triple (3-Shuffle 1-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	39.27%	30.18%	7.27%	-/-	-/-	23.27%	-/-
	Gemini-2.5-Pro	37.78%	22.96%	8.89%	-/-	-/-	30.37%	-/-
	Claude-3.5-Sonnet	29.55%	24.80%	19.26%	-/-	-/-	26.39%	-/-
	Qwen3-VL-235B-Thinking	39.73%	30.14%	6.39%	-/-	-/-	23.74%	-/-
	Qwen2.5-VL-72B-Instruct	27.05%	25.41%	19.67%	-/-	-/-	27.87%	-/-
<13B	Qwen2-VL-2B-Instruct	23.86%	20.75%	24.48%	-/-	-/-	24.07%	6.85%
	DeepSeek-VL2-4.5B	25.75%	24.25%	25.75%	-/-	-/-	24.25%	-/-
	Qwen2.5-VL-7B-Instruct	27.35%	25.38%	18.82%	-/-	-/-	28.45%	-/-
	Qwen3-VL-4B-Instruct	23.16%	23.95%	16.84%	-/-	-/-	21.32%	14.74%
	LLaVA-1.6-Vicuna-7B	23.82%	24.61%	26.57%	-/-	-/-	25.00%	-/-
	Emu3-8B	22.48%	23.06%	24.42%	-/-	-/-	23.45%	6.59%
	Idefics3-8B	24.45%	23.26%	26.24%	-/-	-/-	26.04%	-/-
	Qwen3-VL-8B-Instruct	29.77%	25.75%	14.38%	-/-	-/-	25.08%	5.02%
	InternVL2.5-8B	25.15%	24.76%	23.78%	-/-	-/-	26.32%	-/-
	MiniCPM-V2.6-8B	24.23%	25.58%	24.04%	-/-	-/-	26.15%	-/-
≥13B	Pixtral-12B	25.94%	25.94%	24.06%	-/-	-/-	24.06%	-/-
	LLaVA-1.6-Vicuna-13B	22.31%	25.19%	26.15%	-/-	-/-	26.35%	-/-
Supervised	Qwen2.5-VL-32B	30.24%	28.38%	11.41%	-/-	-/-	29.97%	-/-
	CoPINet	20.85%	26.23%	27.35%	-/-	-/-	25.56%	-/-
	SRAN	24.15%	23.21%	26.04%	-/-	-/-	26.60%	-/-
	MRNet	18.52%	34.07%	29.63%	-/-	-/-	17.78%	-/-
	PredRNet	25.20%	29.20%	29.20%	-/-	-/-	16.40%	-/-
	HCV-ARR	22.96%	26.89%	30.21%	-/-	-/-	19.94%	-/-
SFT	Qwen3-VL-4B-Instruct	22.78%	22.78%	19.62%	-/-	-/-	34.81%	-/-
	Human	40.62%	18.75%	9.38%	-/-	-/-	31.25%	-/-

Table 19. Error analysis for Triple 3-Shuffle 2-Layer configuration on the RAVEN3D Dataset.

	Models	Triple (3-Shuffle 2-Layer)						
		Size	Type	Color	Tier	Number	Position	Invalid
API-based	GPT-4o	33.08%	22.81%	6.08%	20.53%	-/-	17.49%	-/-
	Gemini-2.5-Pro	29.12%	24.21%	10.53%	11.58%	-/-	24.56%	-/-
	Claude-3.5-Sonnet	23.92%	22.94%	12.35%	17.84%	-/-	22.94%	-/-
	Qwen3-VL-235B-Thinking	28.29%	26.29%	7.17%	16.73%	-/-	21.51%	-/-
	Qwen2.5-VL-72B-Instruct	25.35%	21.78%	14.46%	20.40%	-/-	18.02%	-/-
<13B	Qwen2-VL-2B-Instruct	18.76%	19.93%	19.26%	19.10%	-/-	20.44%	2.51%
	DeepSeek-VL2-4.5B	19.29%	19.46%	19.12%	20.30%	-/-	21.83%	-/-
	Qwen2.5-VL-7B-Instruct	22.10%	19.34%	16.94%	19.89%	-/-	21.73%	-/-
	Qwen3-VL-4B-Instruct	20.19%	18.56%	16.94%	16.24%	-/-	19.03%	9.05%
	LLaVA-1.6-Vicuna-7B	17.89%	19.01%	21.57%	21.09%	-/-	20.45%	-/-
	Emu3-8B	16.22%	19.08%	19.71%	17.81%	-/-	22.42%	4.77%
	Idefics3-8B	18.75%	19.79%	20.68%	19.20%	-/-	21.58%	-/-
	Qwen3-VL-8B-Instruct	23.49%	23.00%	12.59%	15.50%	-/-	22.28%	3.15%
	InternVL2.5-8B	20.13%	16.98%	21.07%	20.13%	-/-	21.70%	-/-
	MiniCPM-V2.6-8B	17.73%	19.60%	20.68%	19.91%	-/-	22.08%	-/-
>=13B	Pixtral-12B	22.17%	20.26%	19.46%	18.18%	-/-	19.94%	-/-
	LLaVA-1.6-Vicuna-13B	18.45%	20.92%	20.76%	21.25%	-/-	18.62%	-/-
Supervised	Qwen2.5-VL-32B	23.82%	19.96%	14.16%	18.67%	-/-	23.39%	-/-
	CoPINet	21.56%	22.33%	21.37%	11.83%	-/-	22.90%	-/-
	SRAN	18.44%	19.91%	21.24%	19.47%	-/-	20.94%	-/-
	MRNet	18.86%	27.19%	27.19%	10.53%	-/-	16.23%	-/-
	PredRNet	24.21%	29.47%	23.68%	3.16%	-/-	19.47%	-/-
SFT	HCV-ARR	20.32%	27.09%	31.08%	1.59%	-/-	19.92%	-/-
	Qwen3-VL-4B-Instruct	25.00%	25.86%	17.24%	4.31%	-/-	27.59%	-/-
	Human	48.57%	17.14%	8.57%	8.57%	-/-	17.14%	-/-

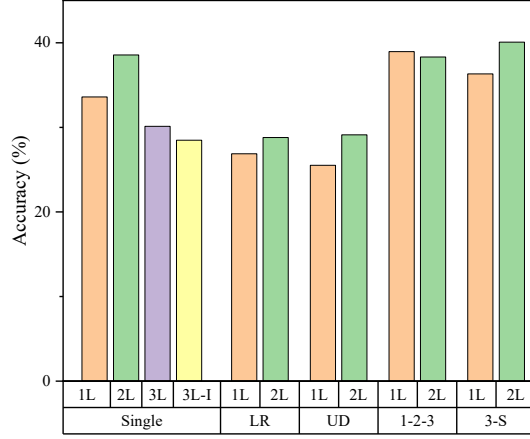


Figure 7. Performance analysis across layer settings for API-based VLMs. 1L/2L/3L use shared attribute sampling across layers; 3L-I uses independent sampling.

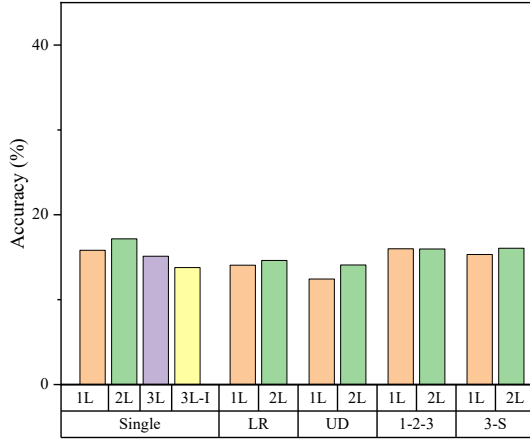


Figure 8. Performance analysis across layer settings for open-source VLMs. 1L/2L/3L use shared attribute sampling across layers; 3L-I uses independent sampling.

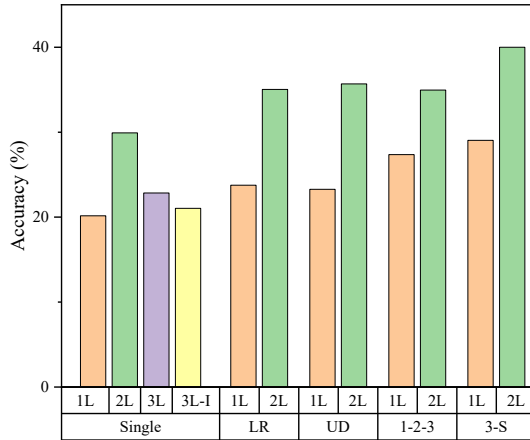


Figure 9. Performance analysis across layer settings for supervised AVR neural networks. 1L/2L/3L use shared attribute sampling across layers; 3L-I uses independent sampling.

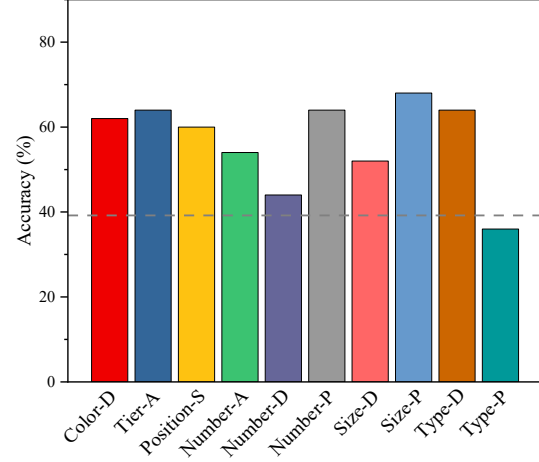


Figure 10. Single-attribute vs. Multi-attribute relation accuracy (GPT-4o). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of GPT-4o on the 2-layer Triple 1-or-2-or-3 setting.

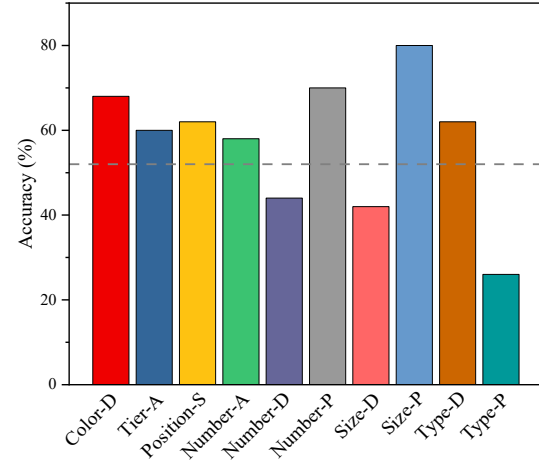


Figure 11. Single-attribute vs. Multi-attribute relation accuracy (Gemini-2.5-Pro). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of Gemini-2.5-Pro on the 2-layer Triple 1-or-2-or-3 setting.

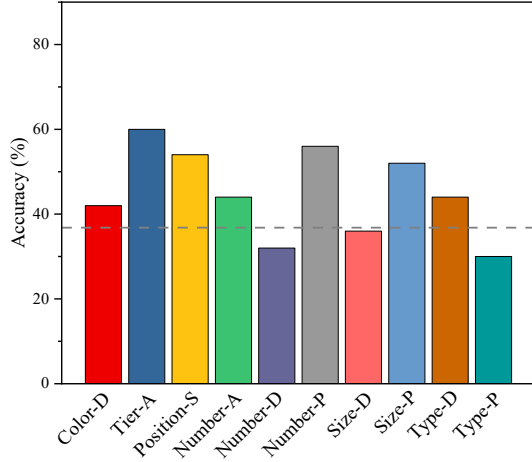


Figure 12. Single-attribute vs. Multi-attribute relation accuracy (Claude-3.5-Sonnet). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of Claude-3.5-Sonnet on the 2-layer Triple 1-or-2-or-3 setting.

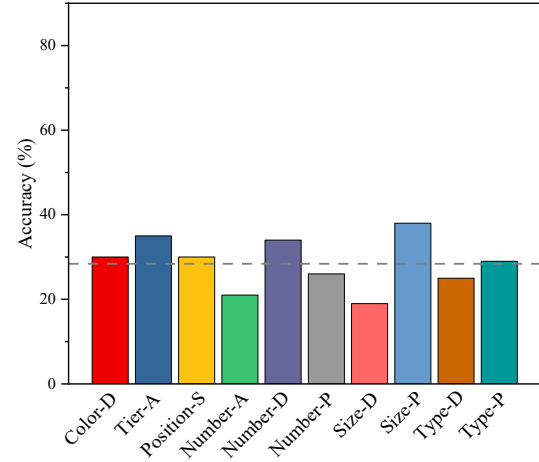


Figure 14. Single-attribute vs. Multi-attribute relation accuracy (CoPINet). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of CoPINet on the 2-layer Triple 1-or-2-or-3 setting.

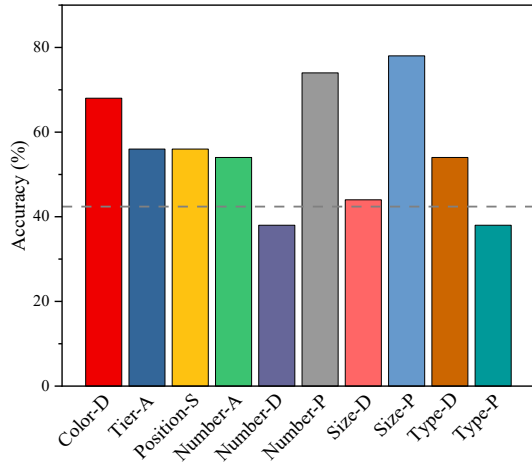


Figure 13. Single-attribute vs. Multi-attribute relation accuracy (Qwen3-VL-235B-Thinking). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of Qwen3-VL-235B-Thinking on the 2-layer Triple 1-or-2-or-3 setting.

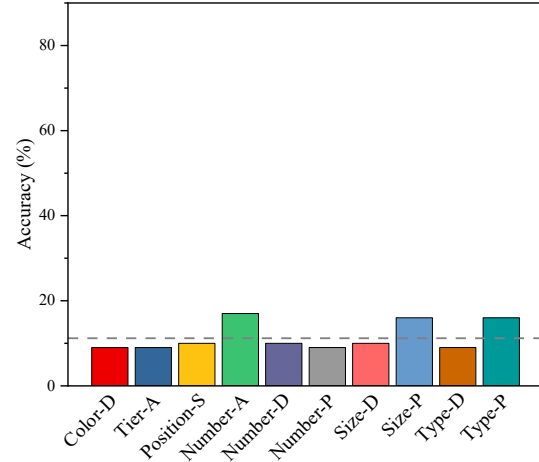


Figure 15. Single-attribute vs. Multi-attribute relation accuracy (SRAN). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of SRAN on the 2-layer Triple 1-or-2-or-3 setting.

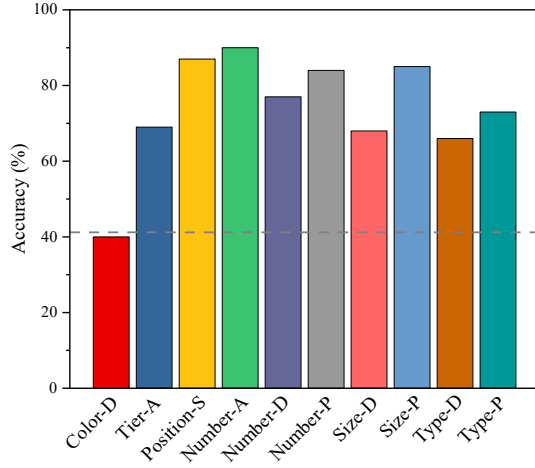


Figure 16. Single-attribute vs. Multi-attribute relation accuracy (MRNet). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of MRNet on the 2-layer Triple 1-or-2-or-3 setting.

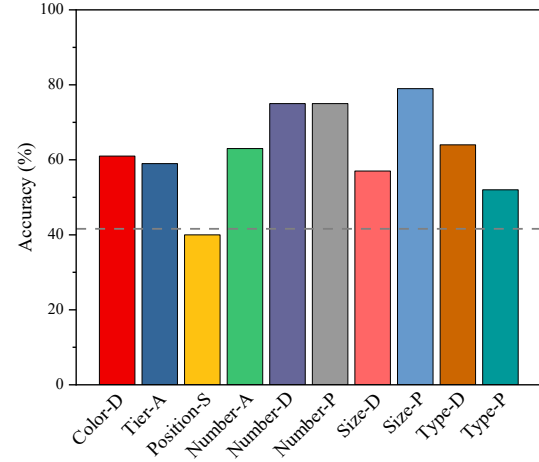


Figure 18. Single-attribute vs. Multi-attribute relation accuracy (HCV-ARR). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of HCV-ARR on the 2-layer Triple 1-or-2-or-3 setting.

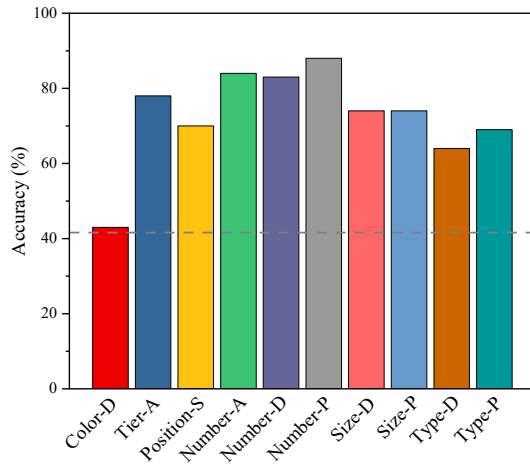


Figure 17. Single-attribute vs. Multi-attribute relation accuracy (PredRNet). A, D, P, and S denote the Arithmetic, Distribute-Three, Progression, and Shuffle relations, respectively. The gray dashed line indicates the accuracy of PredRNet on the 2-layer Triple 1-or-2-or-3 setting.

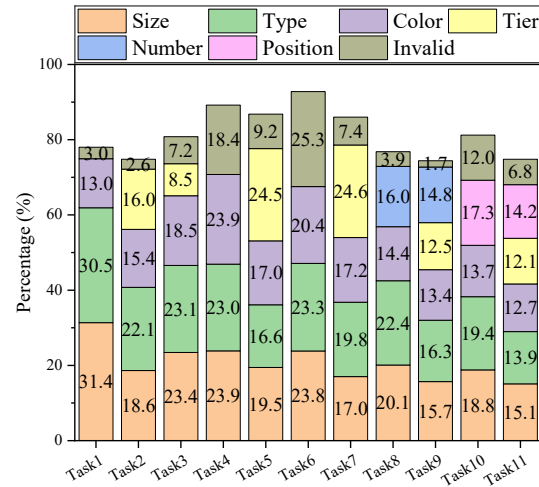


Figure 19. Error analysis of Qwen3-VL-4B-Instruct. Task 1 to Task 11 correspond to the subtasks of Single (1-layer, 2-layer and 3-layers), Double (1-layer and 2-layer of Left-Right and Up-Down) and Triple (1-layer and 2-layer of 1-or-2-or-3 and 3-Shuffle) respectively.

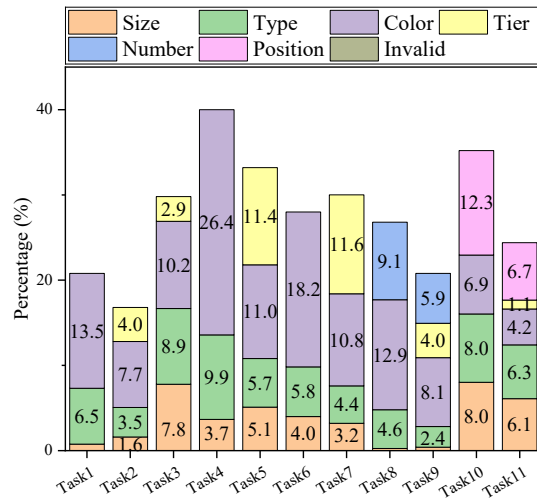


Figure 20. Error analysis of fine-tuned Qwen3-VL-4B-Instruct. Task 1 to Task 11 correspond to the subtasks of Single (1-layer, 2-layer and 3-layers), Double (1-layer and 2-layer of Left-Right and Up-Down) and Triple (1-layer and 2-layer of 1-or-2-or-3 and 3-Shuffle) respectively.

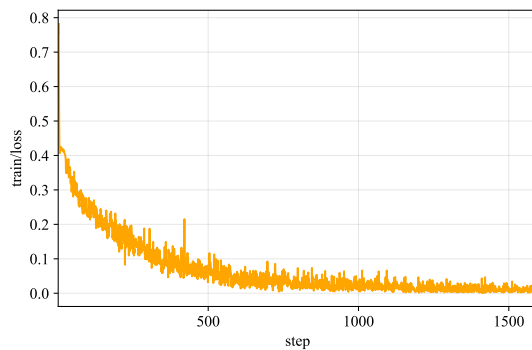


Figure 21. Training loss curves for fine-tuning Qwen3-VL-4B-Instruct.

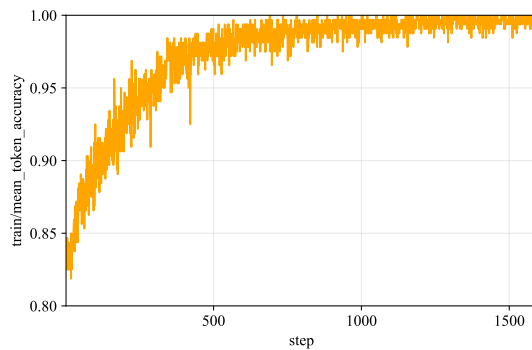


Figure 22. Mean token accuracy across epochs during fine-tuning of Qwen3-VL-4B-Instruct.